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Principles of Macroeconomics

Lecture 6: The monetary system

Prof. Dr. Jan-Egbert Sturm

Evidence from PoMo HS25 students:
Job search influenced by more than salary
Some students are willing to earn less for jobs that fit their values

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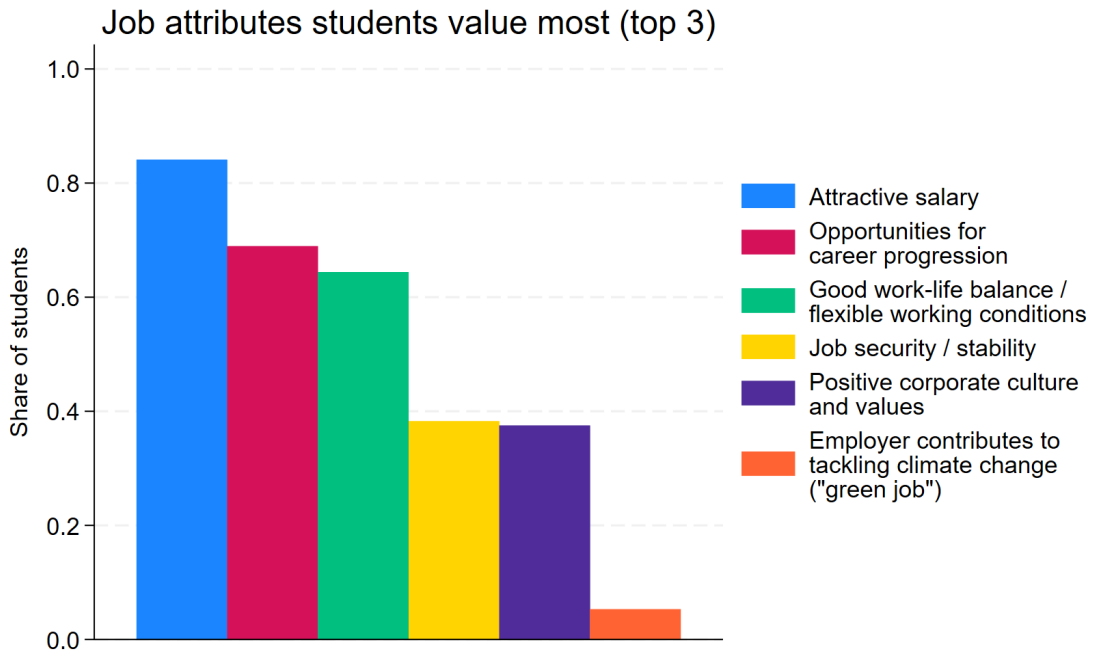
«Attractive salary» is the most frequently chosen job attribute (Note: this does not hold for 1 in 6 students)



The average student would give up 7% of their pay for a “green job”

Schedule

Date	Lecture	Topic	Chapters
16/09/2025	1	Introduction	1, 2
23/09/2025	2	National Accounting	20
30/09/2025	3	Production and Growth	21
07/10/2025	4	Unemployment and the labour market	22
14/10/2025	5	Saving, investment and the financial system	23
21/10/2025	6	The Monetary System	24
28/10/2025	7	International Trade	3,15,17
04/11/2025	8	Open-Economy Macroeconomics	25
11/11/2025	9	Business Cycles	26
18/11/2025	10	Keynesian Economics and IS-LM Analysis	27
25/11/2025	11	Aggregate Demand and Aggregate Supply	28
02/12/2025	12	Monetary and Fiscal Policy	29
09/12/2025	13	Phillips Curve / Supply-side policies	30,31
16/12/2025	14	Exam preparation	

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Principles of Macroeconomics

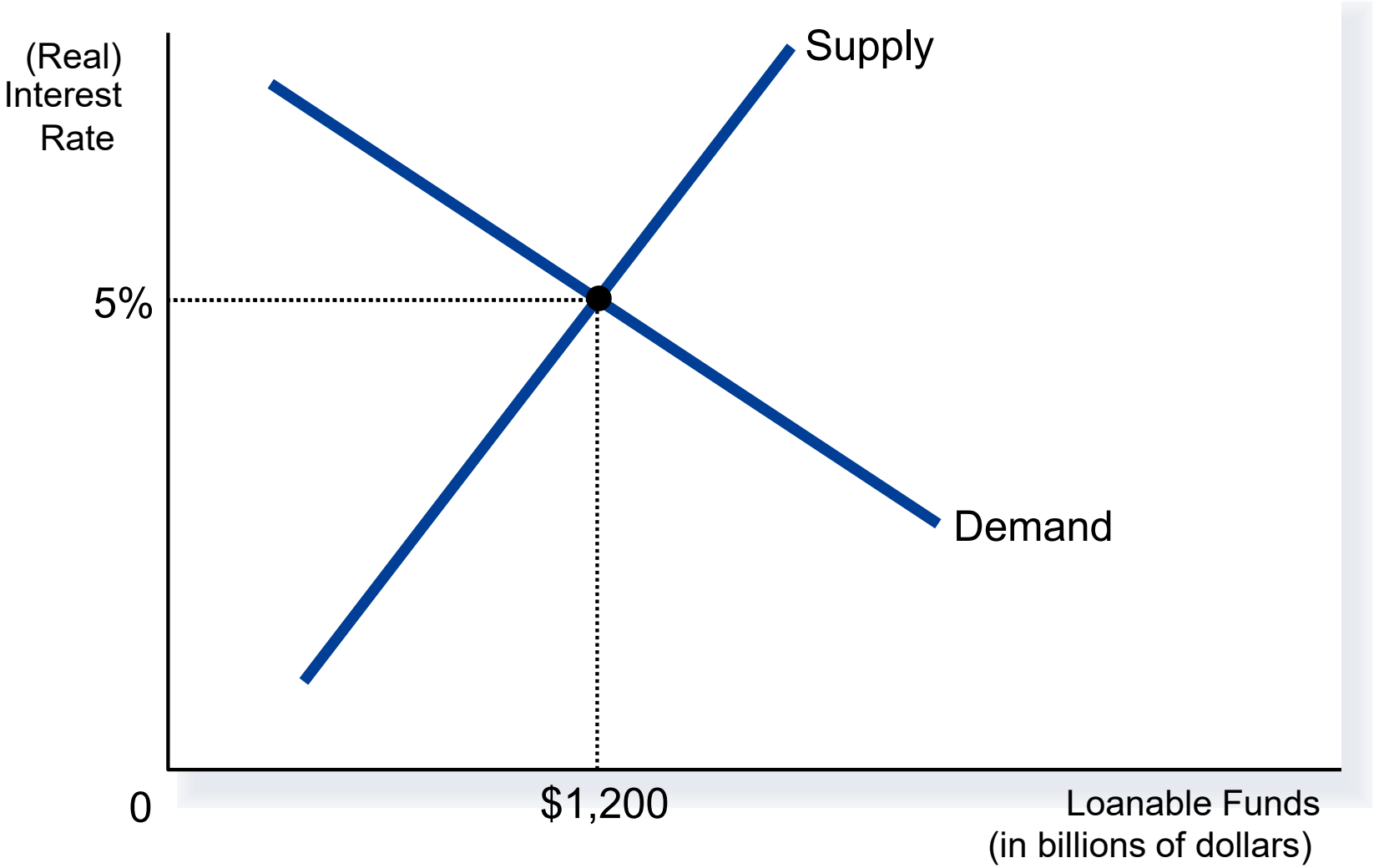
Lecture 5: Saving, investment and the financial system

Prof. Dr. Jan-Egbert Sturm

The Market for Loanable Funds

- The savings and investment rate determines the growth rate and long-run level of GDP
- Saving and investment are coordinated on the financial market
- We will assume for simplicity that the economy has only **one** financial market
  - This ***market for loanable funds*** is the market in which those who want to save supply funds and those who want to borrow to invest demand funds
  - Loanable funds refers to all income that people have chosen to save and lend out (S), rather than use for their own consumption
- The price for a loanable fund is the (real) interest rate
  - It reflects the amount borrowers pay for loans and lenders receive on their saving
- Financial markets link the present to the future and work much like other markets in the economy
  - A decision to save is a postponement of consumption in the present to consumption in the future
    - The intertemporal substitution effect is the rate at which economic actors respond to changes in the interest rate by changing savings and investment decisions
  - The equilibrium of the supply (S) and demand (I) for loanable funds determines the real interest rate

The Market for Loanable Funds - Graphical Illustration



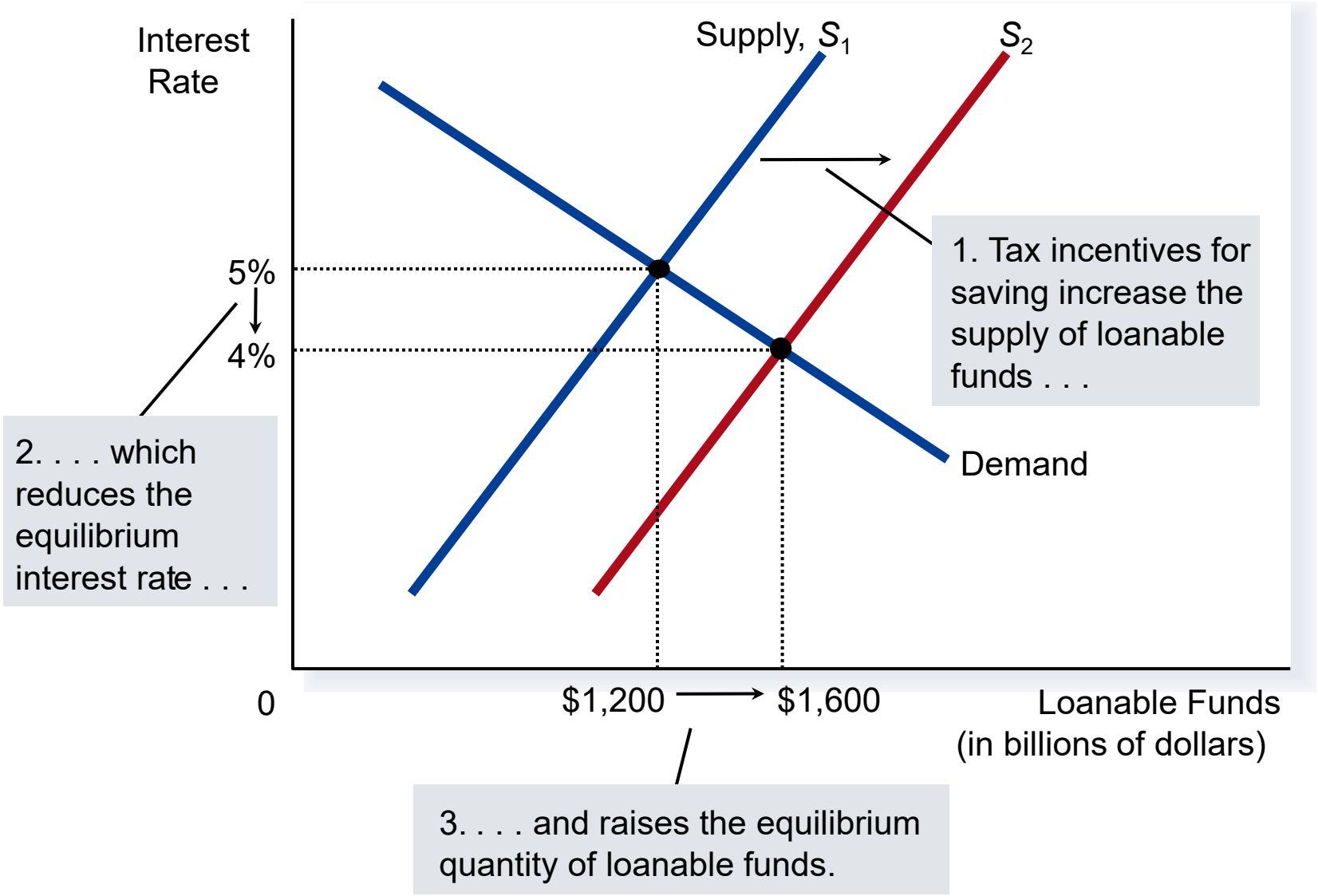
Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
3. Government budget deficits and surpluses

Government Policies to affect Savings and Investment

1. Taxes and saving
   - Taxes on interest income substantially reduce the future payoff from current saving and, as a result, reduce the incentive to save
   - A tax decrease increases the incentive for households to save at any given interest rate
     - The supply of loanable funds curve shifts right
     - The equilibrium interest rate decreases
     - The equilibrium quantity of loanable funds increases
2. Taxes and investment
3. Government budget deficits and surpluses

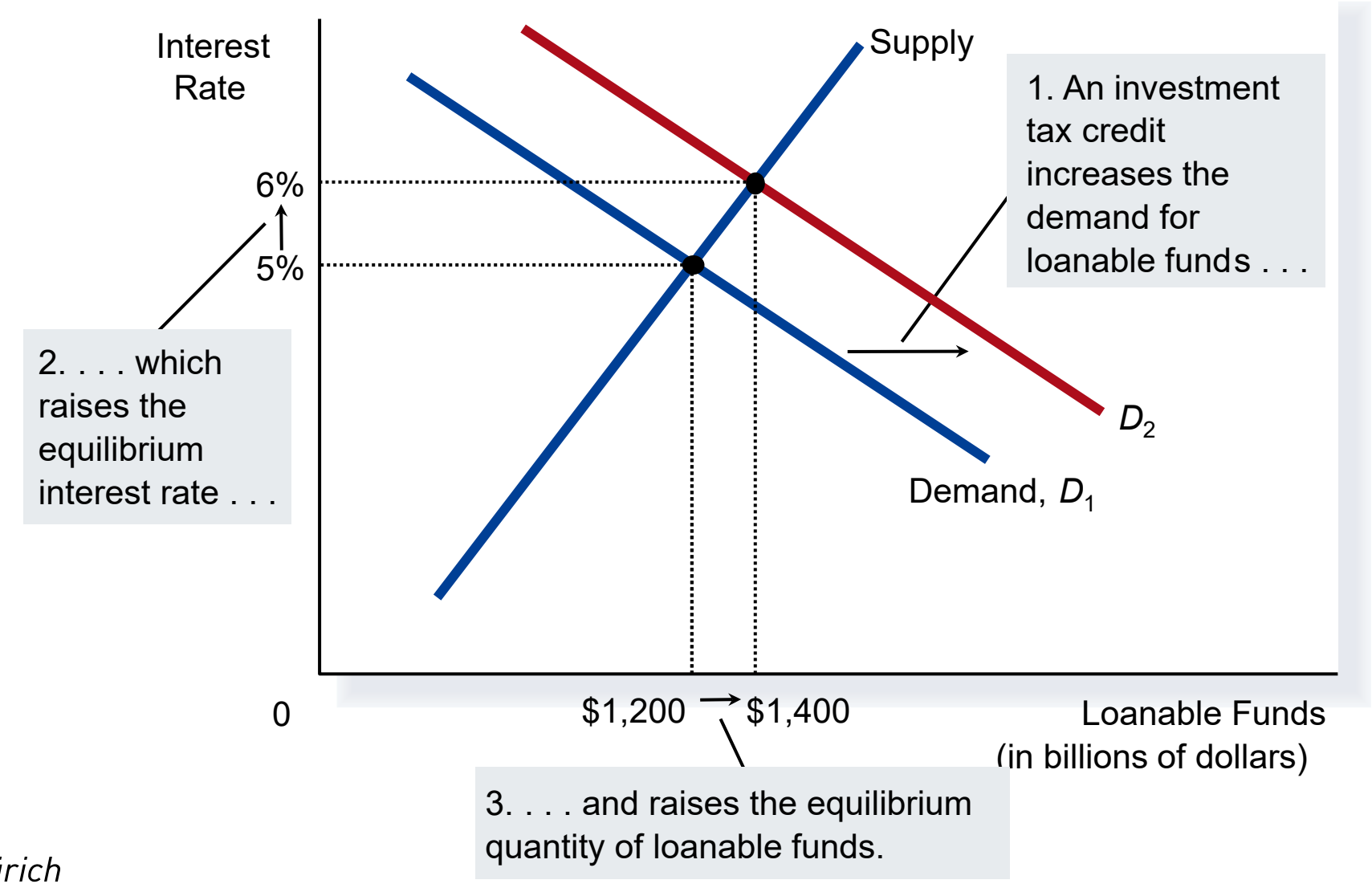
An Increase in the Supply of Loanable Funds



Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
   - An investment tax credit
     - increases the incentive to borrow
     - increases the demand for loanable funds
     - shifts the demand curve to the right
     - results in a higher interest rate and a greater quantity saved
3. Government budget deficits and surpluses

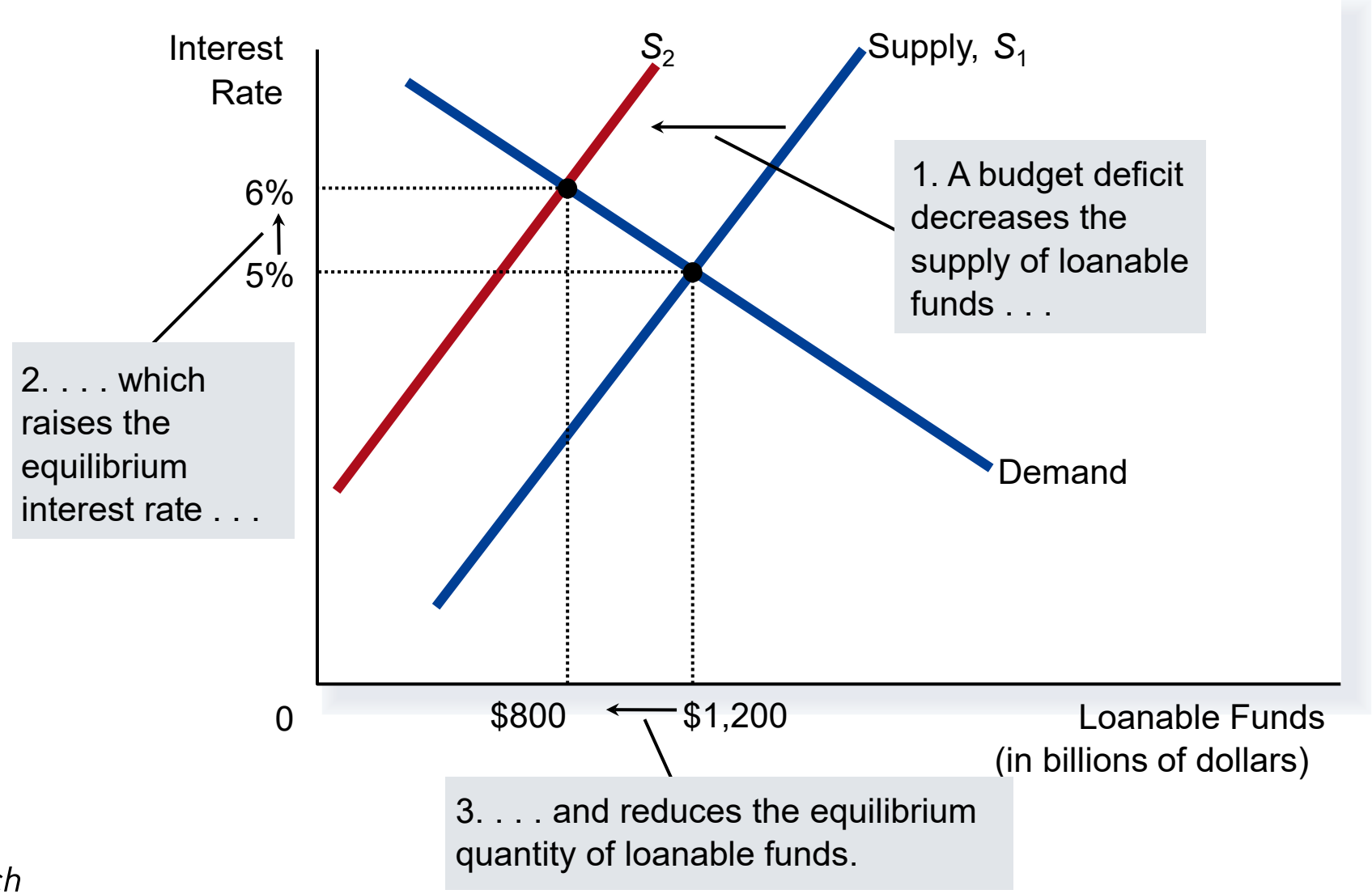
Investment Incentives Increase the Demand for Loanable Funds



Government Policies to affect Savings and Investment

1. Taxes and saving
2. Taxes and investment
3. Government budget deficits and surpluses
   - Government borrowing to finance its budget deficit reduces the supply of loanable funds available to finance investment by households and firms
   - A budget deficit decreases the supply of loanable funds
     - Shifts the supply curve to the left
   - Increases the equilibrium interest rate and reduces the equilibrium quantity of loanable funds
   - The associated fall in investment is referred to as **crowding out**
     - The deficit borrowing crowds out private borrowers who are trying to finance investments

The Effect of a Government Budget Deficit



Summary

- The financial system is made up of financial institutions such as the bond market, the stock market, banks, and mutual funds
- All these institutions act to direct the resources of households who want to save some of their income into the hands of households and firms who want to borrow
- National income accounting identities reveal some important relationships among macroeconomic variables
- In particular, in a closed economy, national saving must equal investment
- The savings and investment rate determines the growth rate and long-run level of GDP and consumption
- Financial institutions attempt to match one person's saving with another person's investment
- The interest rate is determined by the supply and demand for loanable funds
- The supply of loanable funds comes from households who want to save some of their income
- The demand for loanable funds comes from households and firms who want to borrow for investment
- National saving equals private saving plus public saving
- A government budget deficit represents negative public saving and, therefore, reduces national saving and the supply of loanable funds
- When a government budget deficit crowds out investment, it reduces the growth of productivity and GDP

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Principles of Macroeconomics

Lecture 6: The monetary system

Prof. Dr. Jan-Egbert Sturm

The Meaning of Money

- ***Bartering*** is the exchange of one good for another.
  - Bartering requires a double coincidence of wants.

- ***Money*** is the set of assets in an economy that people regularly use to buy goods and services from other people.

The Three Functions of Money

1. Medium of Exchange
   - A ***medium of exchange*** is an item that buyers give to sellers when they want to purchase goods and services.
   - A medium of exchange is anything that is readily acceptable as payment.
2. Unit of Account
   - A ***unit of account*** is the yardstick people use to post prices and record debts.
3. Store of Value
   - A ***store of value*** is an item that people can use to transfer purchasing power from the present to the future.

Other important dimensions

The Liquidity of an Asset

- ***Liquidity*** is the ease with which an asset can be converted into the economy's medium of exchange.

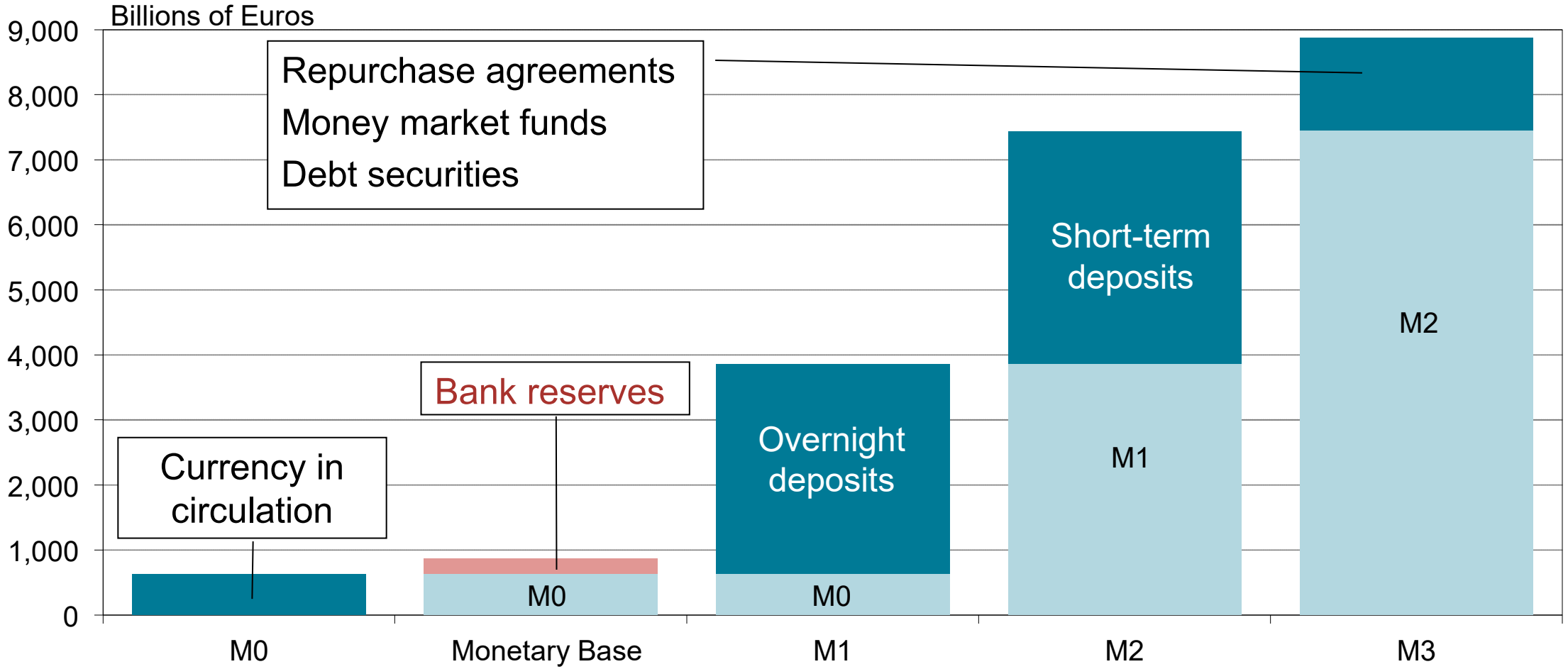
The Kinds of Money

- ***Commodity money*** takes the form of a commodity with intrinsic value.
  - Examples: Gold, silver, cigarettes.
- ***Fiat money*** is used as money because of government decree.
  - It does not have intrinsic value.
  - Examples: Coins, currency, check deposits.

Money in the Economy

- ***Currency*** is the paper bills and coins in the hands of the public.
- ***Demand deposits*** are balances in bank accounts that depositors can access on demand.

Money supply measures, Euro area, 01/01/2008



Source: ECB

Money supply measures (before the Financial Crisis)

Symbol	Assets included	Switzerland, 1.1.08 Amount (in blns)	Switzerland, 1.1.08 ratio to MB	Euro area, 1.1.08 Amount (in blns)	Euro area, 1.1.08 ratio to MB
MB	Currency (M0) + Bank reserves	SFr. 47		€ 842	
M1	M0 + demand deposits, travelers' checks, other checkable deposits	SFr. 271	5.8	€ 3,839	4.6
M2	M1 + small time deposits, savings deposits, money market deposit accounts	SFr. 458	9.8	€ 7,355	8.7
M3	M2 + large time deposits, repurchase agreements, institutional money market mutual fund balances, debt securities	SFr. 641	13.7	€ 8,650	10.3

Sources: ECB, SNB

Functions of a Central Bank

- A ***central bank*** is an institution designed to oversee the banking system and regulate the quantity of money in the economy.
- Whenever an economy relies on fiat money, there must be some agency that regulates the system.

- Two Primary Functions of a Central Bank
  1. Act as a banker's bank, making loans to banks and as a ***lender of last resort***.
     - assuring financial stability
  2. Conducts ***monetary policy***
     - by controlling the money supply
     - by controlling the internal value of the currency (price stability)
     - by controlling the external value of the currency (exchange rate)

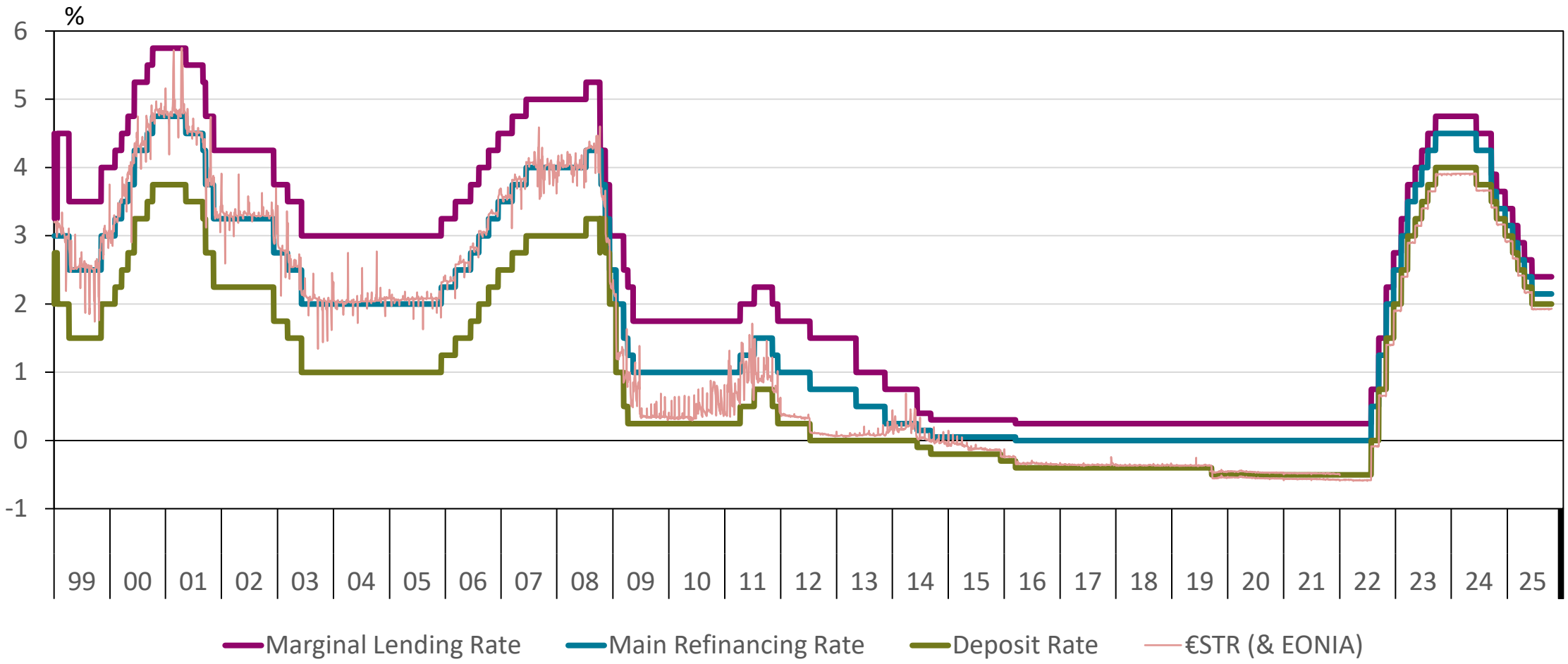
The European Central Bank (ECB) and the Eurosystem

- The ***European Central Bank*** is the overall central bank of the 20 countries comprising the European Monetary Union.
- The ECB was officially created on 1 June 1998 and is located in Frankfurt.
- It came into being because 11 countries of the European Union had decided that they wished to enter European Monetary Union and use the same currency.
- The primary objective of the ECB is to promote ***price stability*** throughout the euro area.
- An important feature of the ECB and the Eurosystem is its independence.
- The ***Eurosystem*** is the system made up of the ECB plus the national central banks of the 20 countries comprising the ***European Monetary Union***.

The Central Bank's (conventional) Tools of Monetary Control

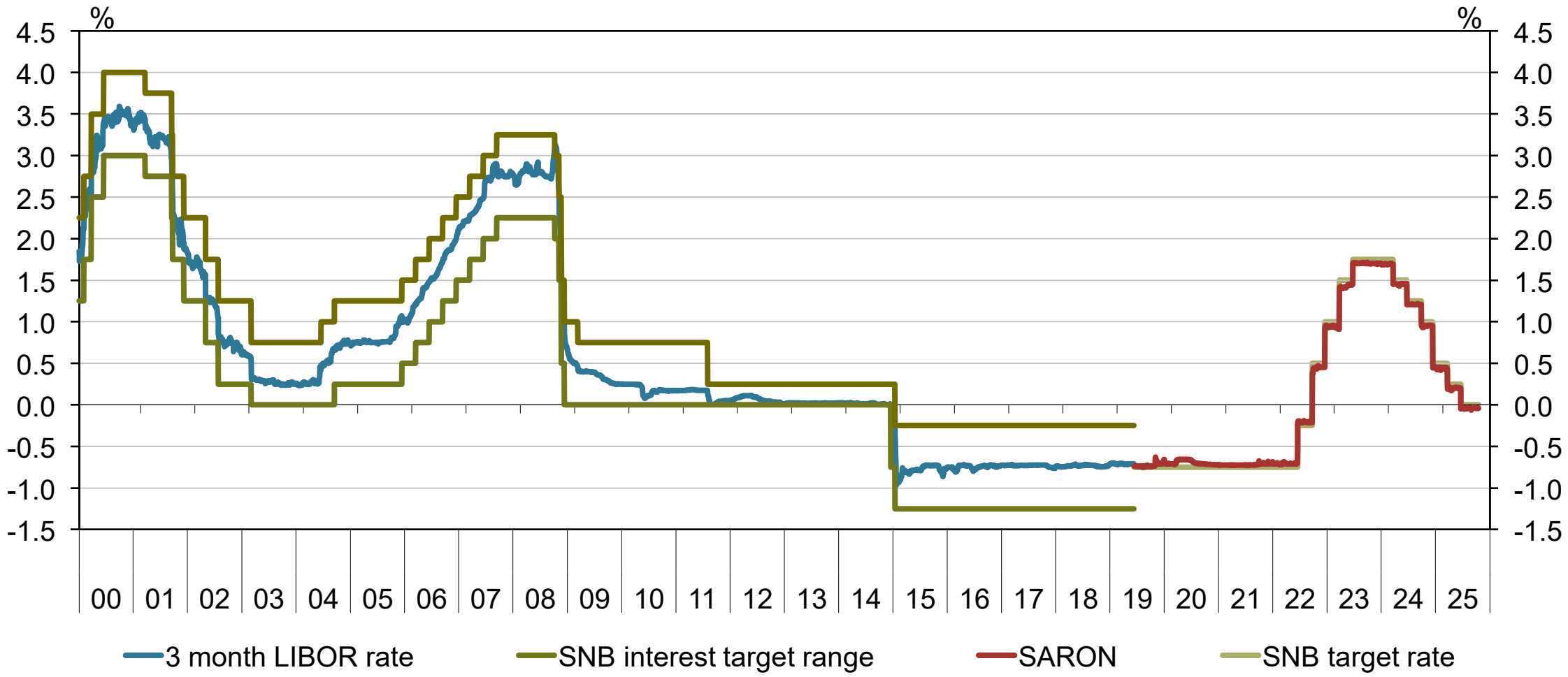
- Reserve Requirements
  - ***Reserve requirements*** are regulations on the minimum amount of reserves banks must hold against deposits
    - Increasing (decreasing) the reserve requirement decreases (increases) the money supply.
- Open-Market Operations (in the ECB case using the *Main Refinancing Rate* as "price")
  - The CB conducts ***open-market operations*** when it buys government bonds from or sells government bonds to the public:
    - When the CB buys (sells) government bonds, the money supply increases (decreases)
- The Discount Rate (ECB terminology: the *Marginal Lending Rate*)
  - The ***discount rate*** is the interest rate the CB charges banks for loans
    - Increasing (decreasing) the discount rate decreases (increases) the money supply
- The ***deposit rate*** is the interest rate the CB pays banks for (some of the) deposits they hold at the CB
    - Increasing (decreasing) the deposit rate decreases (increases) the money supply
- Normally, CB interest rates move hand in hand and thereby affect money market rates (like the €STR)

Monetary policy rates in the Euro Area



Source: ECB

Monetary policy rates in Switzerland

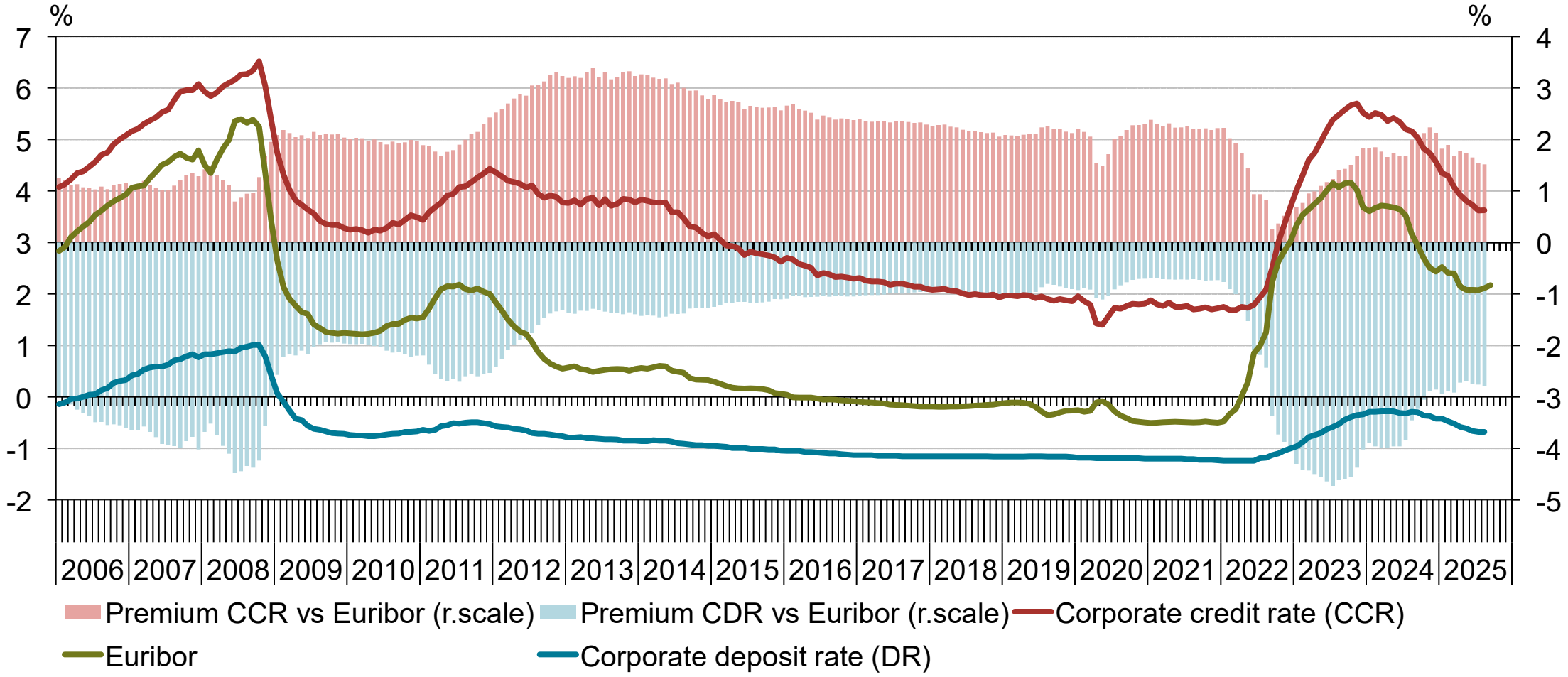


Source: SNB

How Banks Make a Profit

- Most banks make profits by accepting deposits and making loans.
  - The difference between the average interest a bank earns on its assets and the average interest rate paid on its liabilities is termed the spread.
  - Banks hold a fraction of the money deposited as reserves and lend out the rest to make their profit.
    - Note that banks operating under Islamic Sharia principles make profits from the sharing of risk and reward between lenders and borrowers.

Different interest rates in the Euro Area



Source: ECB

Problems in Controlling the Money Supply

- The Central Bank's control of the money supply is not precise.

- The Central Bank must wrestle with two problems that arise due to fractional-reserve banking.
  1. The Central Bank does not control the amount of money that households choose to hold as deposits in banks.
  2. The Central Bank does not control the amount of money that bankers choose to lend and/or the private sector chooses to borrow.

Banks and the Money Supply

- Banks can influence the quantity of demand deposits in the economy and the money supply.
- ***Reserves*** are deposits that banks have received but have not been lend out.
- In a ***fractional-reserve banking*** system, banks hold a fraction of the money deposited as reserves and lend out the rest.
- The ***reserve ratio*** is the fraction of deposits that banks hold as reserves.

- As we will see, when a bank makes a loan from its reserves, the money supply increases.
- The money supply is affected by the amount deposited in banks and the amount that banks lend out.
  - Deposits into a bank are recorded as liabilities.
  - The fraction of total deposits that a bank has to keep as reserves is called the reserve ratio.
  - Loans become an asset to the bank.

Banking Money Creation in a Fractional-Reserve System

First National Bank

Assets	Liabilities
Reserves $10.00	Deposits $100.00
Loans $90.00	
Total Assets $100.00	Total Liabilities $100.00

- This T-Account shows a bank that…
  - accepts deposits,
  - keeps a portion as reserves,
  - and lends out the rest
- Let us assume a reserve ratio of 10%
- From a situation without this bank (i.e. households holding $100 in cash instead of deposits), introducing this bank and using a 10% reserve ratio will increase the money supply (M1, M2, M3) by $90 (to $190)
- M0 (actually the *monetary base*) stays at $100=$90+$10

Money Creation with Fractional-Reserve Banking

- When one bank loans money, that money is generally deposited into another bank.
- This creates more deposits and more reserves to be lent out.
- When a bank makes a loan from its reserves, the money supply increases.

The Money Multiplier

- How much money is eventually created by the new deposit in this economy?
- The ***money multiplier*** is the amount of money the banking system generates with each dollar of reserves.

The Money Multiplier

First National Bank

Assets		Liabilities	
Reserves	$10.00	Deposits	$100.00
Loans	$90.00		
Total Assets	$100.00	Total Liabilities	$100.00

Second National Bank

Assets		Liabilities	
Reserves	$9.00	Deposits	$90.00
Loans	$81.00		
Total Assets	$90.00	Total Liabilities	$90.00

Increase in the Money Supply (M1) = $81.00!

The Money Multiplier

Original deposit = $100.00

- 1st Natl. Lending = 90.00 (=.9 x $100.00)
- 2nd Natl. Lending = 81.00 (=.9 x $ 90.00)
- 3rd Natl. Lending = 72.90 (=.9 x $ 81.00)
- … and on until there are just pennies left to lend!
- Total money created by this $100.00 deposit in the banking sector is $900.00 (= 1/0.10 x $100.00 - $100.00)
  - Total money in the system is $1’000.00
- The money multiplier is the reciprocal of the reserve ratio: **Money Multiplier = 1/R**
- Example:
  - With a reserve requirement of 20%, R = 20% or .2:
    - The money multiplier is 1/.2 = 5

Remember: Money supply measures (before the Financial Crisis)

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Sources: ECB, SNB

Macroprudential Policy

- *Systemic risk* refers to the risk of failure across the whole of the financial sector rather than just to one or two institutions
- *Macroprudential policy* limits the risk across the financial sector by focusing on improving ‘prudential’ standards of operation that enhance stability
  - Banks must be able to respond to both defaults (*credit risk*) and increased withdrawals (*liquidity risk*)
  - The ratio of this capital in relation to the rest of the bank’s assets is seen as a measure of the financial strength of a bank. It effectively acts as a cushion against financial shocks

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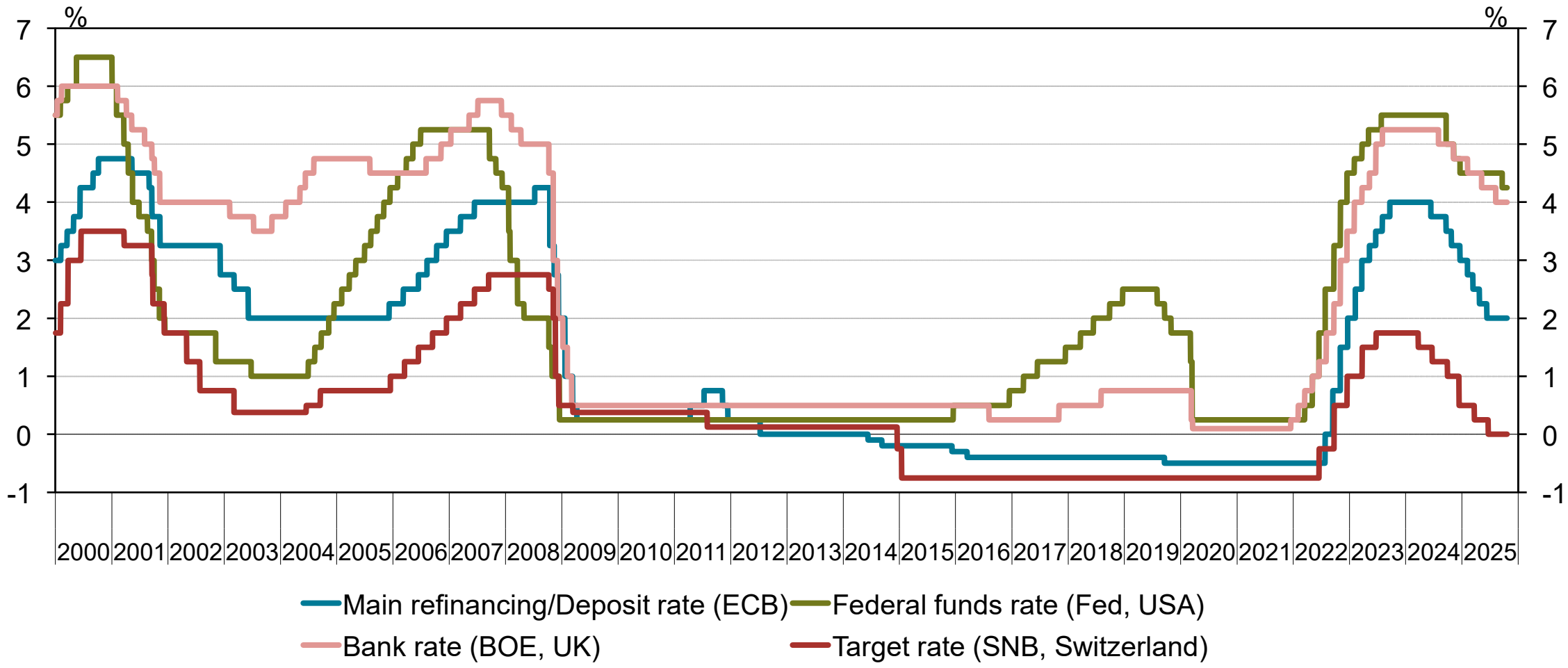
The Great Financial Crisis and beyond…

Effective Lower Bound of Interest Rate Policies

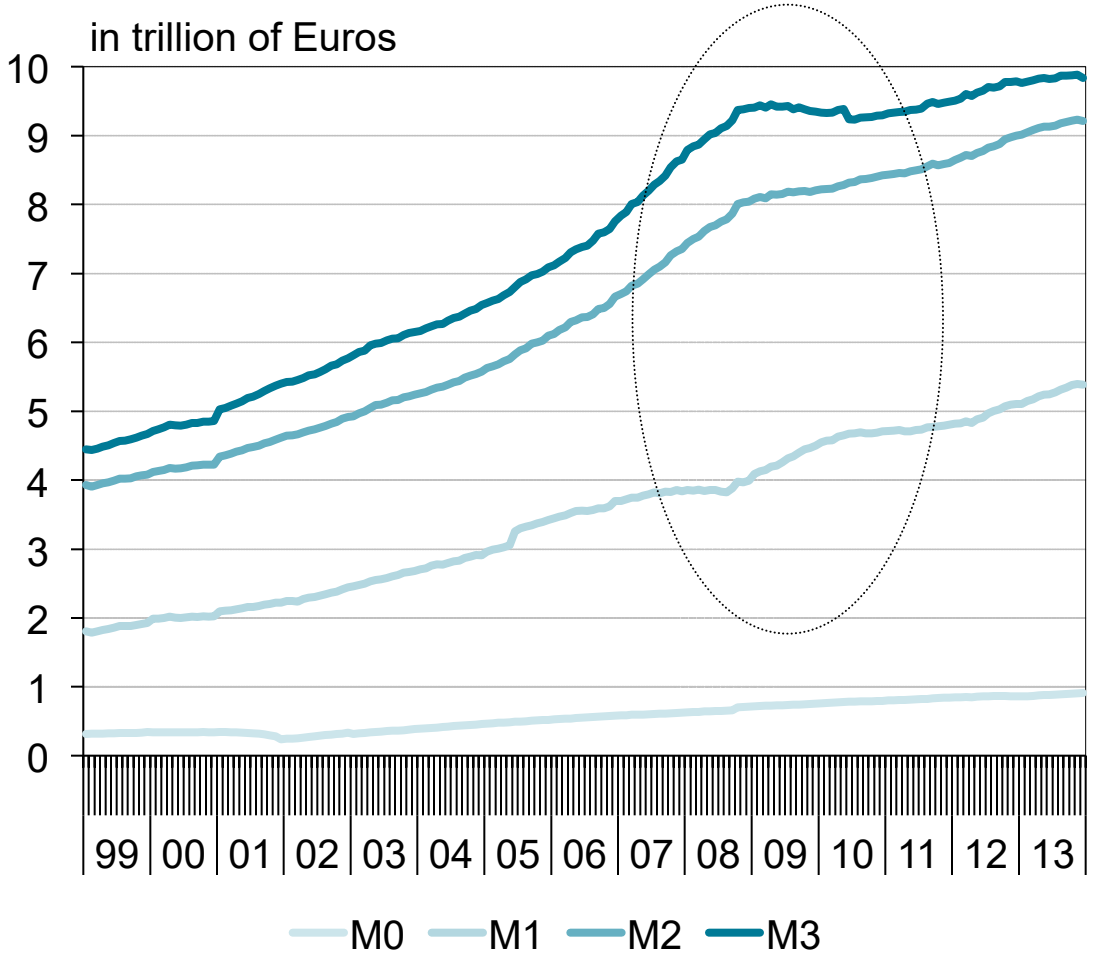


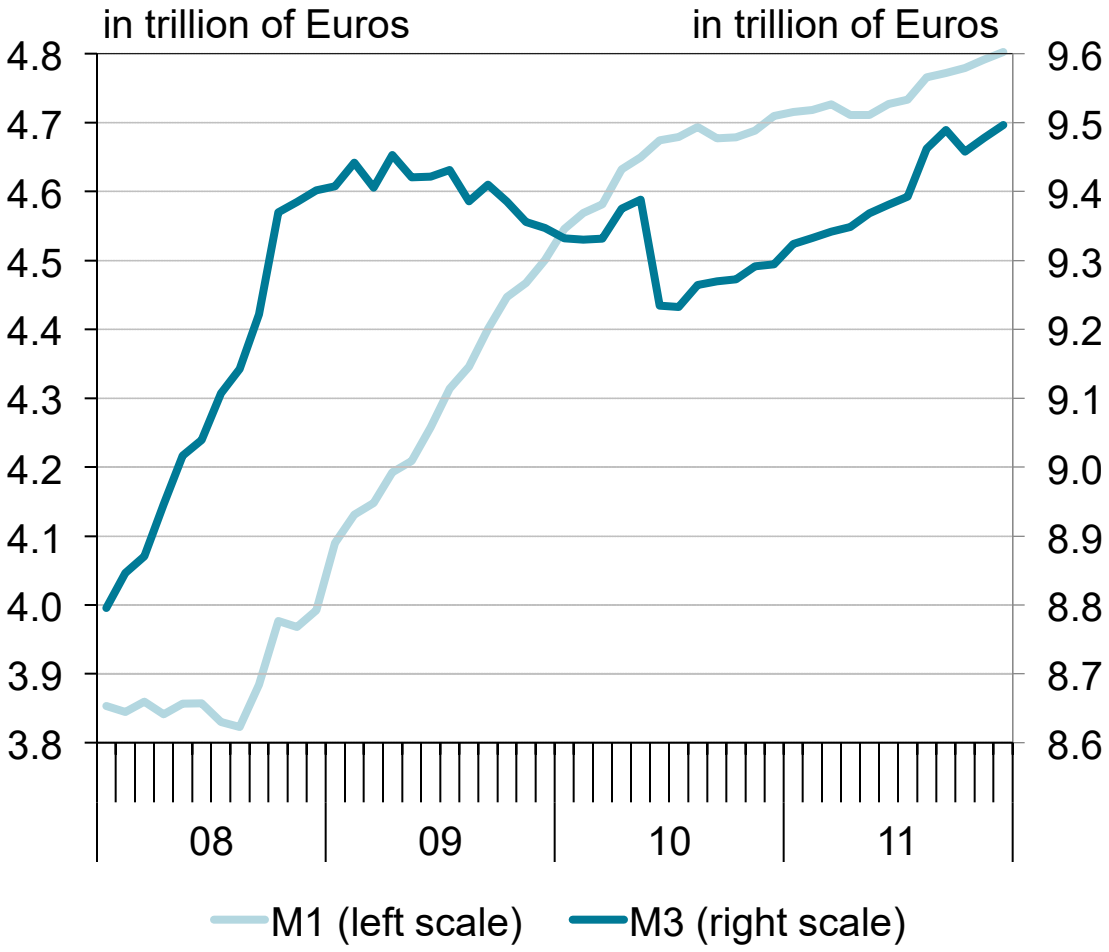
Sources: ECB, Fed, BOE, SNB

Quantitative Easing (Unconventional Policy Instrument)

- Having exhausted the use of lowering the price of money (i.e. lowering the interest rate) during the financial crisis of 2007–09, central banks used ***quantitative easing*** to try and support the economy
- The central bank injects money into the system
  - It buys assets (like government and corporate bonds) from private sector institutions such as banks, pension and insurance companies
  - So the reserves of the private sector institutions increase
  - In theory these institutions should have more money to lend

Money supply in the euro area during the Great Financial Crisis





Source: ECB

Measures for monetary aggregates, 2008 vs. today

Symbol	Assets included	Switzerland 01/01/08 Amount (in blns)	Switzerland 01/01/08 ratio to MB	Switzerland 01/09/25 Amount (in blns)	Switzerland 01/09/25 ratio to MB	Euro Area 01/01/08 Amount (in blns)	Euro Area 01/01/08 ratio to MB	Euro Area 01/09/25 Amount (in blns)	Euro Area 01/09/25 ratio to MB	CH Avg. annual growth rate	EA Avg. annual growth rate
MB	Currency + Bank reserves	SFr. 47		SFr. 511		€ 842		€ 4,430		14.5%	9.8%
M2	M1 + small time deposits, savings deposits, money market deposit accounts	SFr. 458	9.8	SFr. 1,081	2.1	€ 7,356	8.7	€ 15,743	3.6	5.0%	4.4%
M3	M2 + large time deposits, repurchase agreements, institutional money market mutual fund balances, debt securities	SFr. 641	13.7	SFr. 1,197	2.3	€ 8,648	10.3	€ 16,912	3.8	3.6%	3.9%

Sources: SNB, ECB

Money supply measures, Euro area

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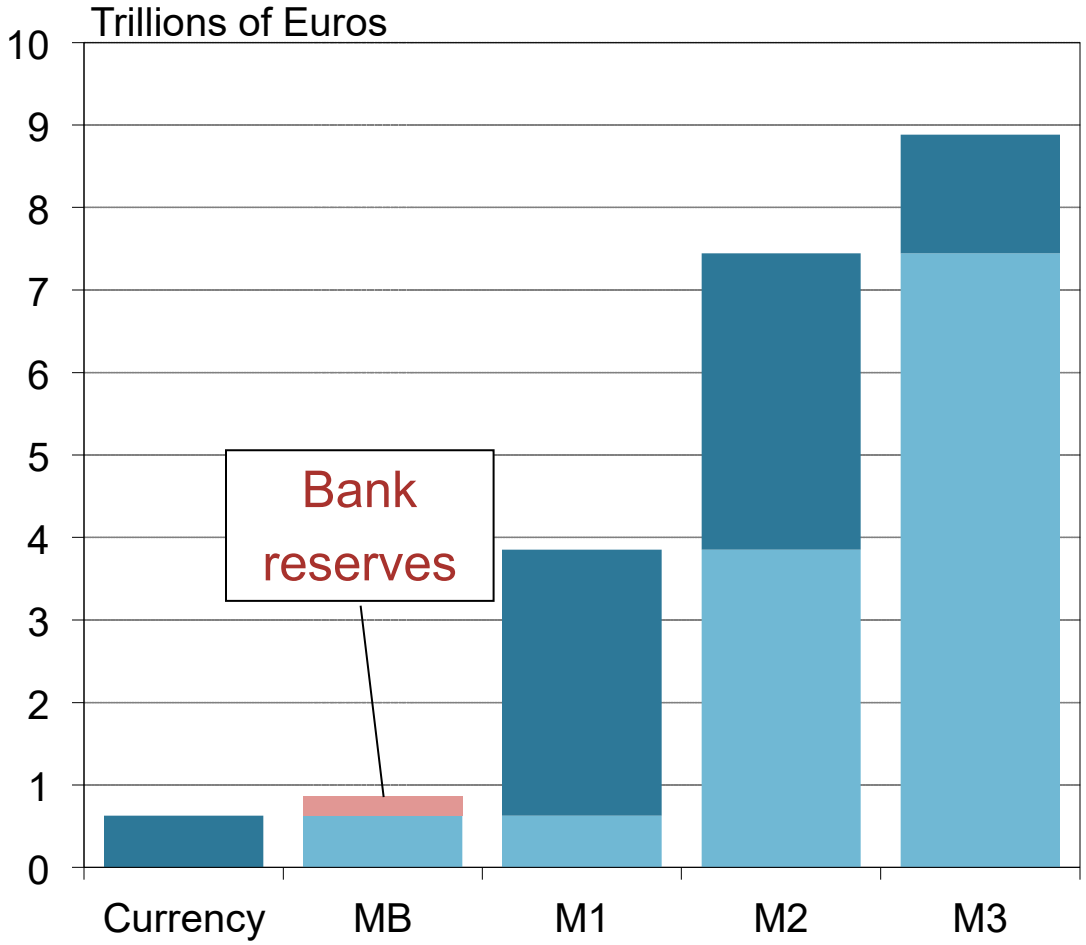
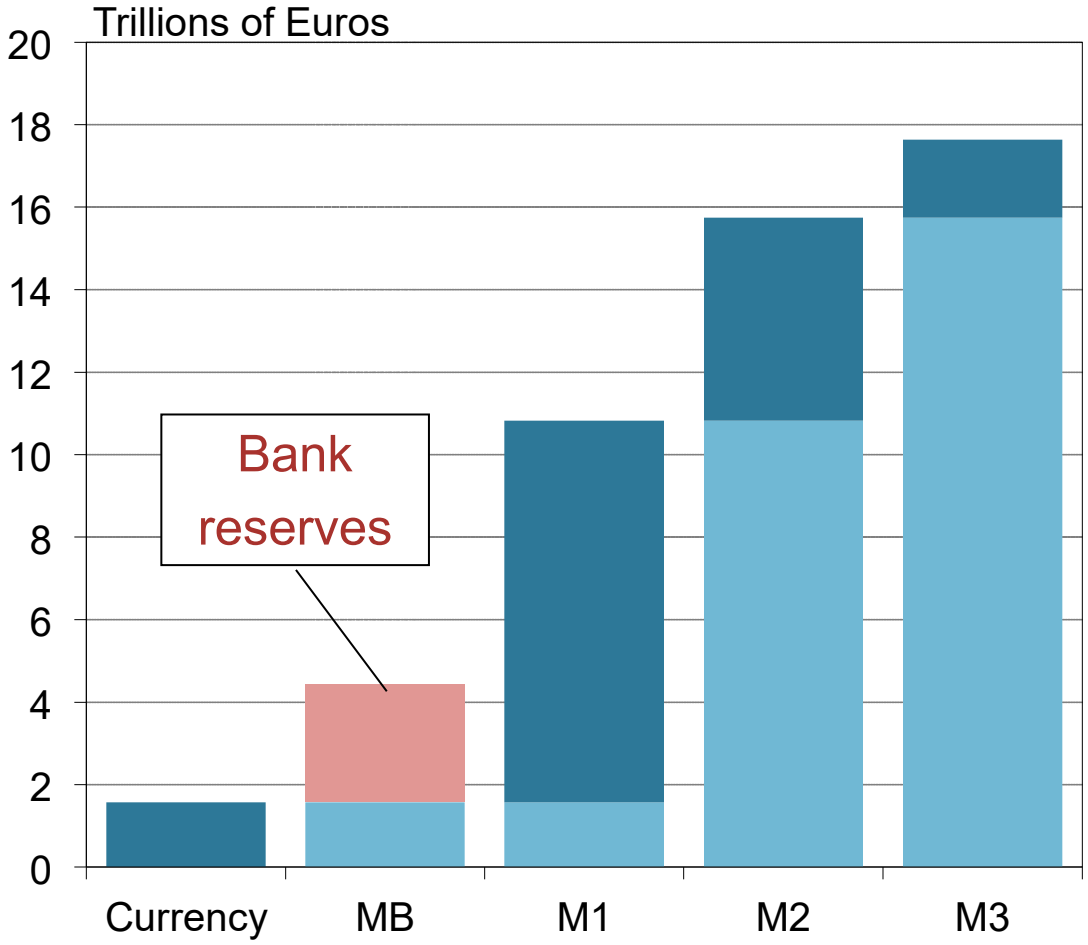
01/01/2008

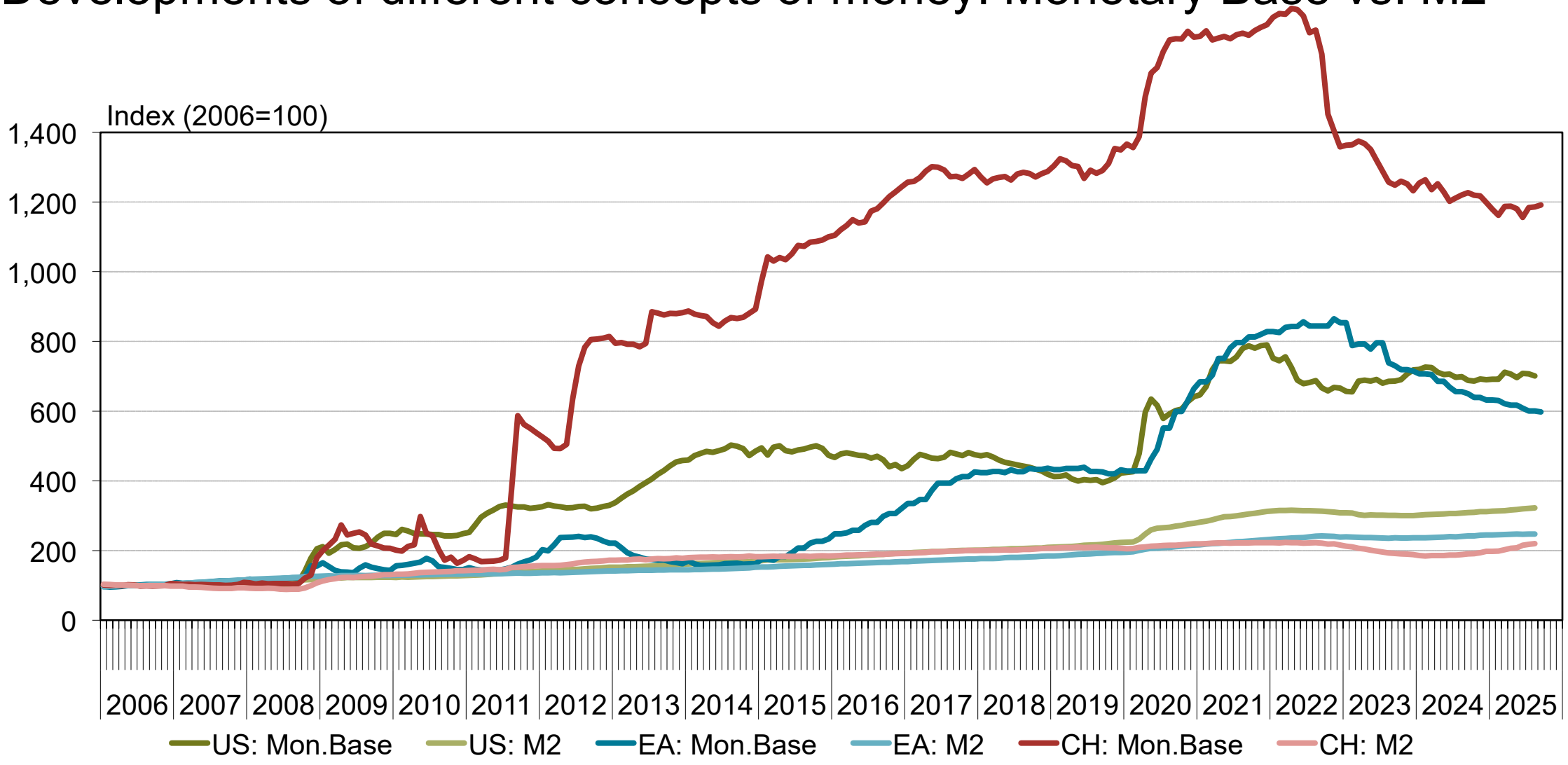


01/09/2024



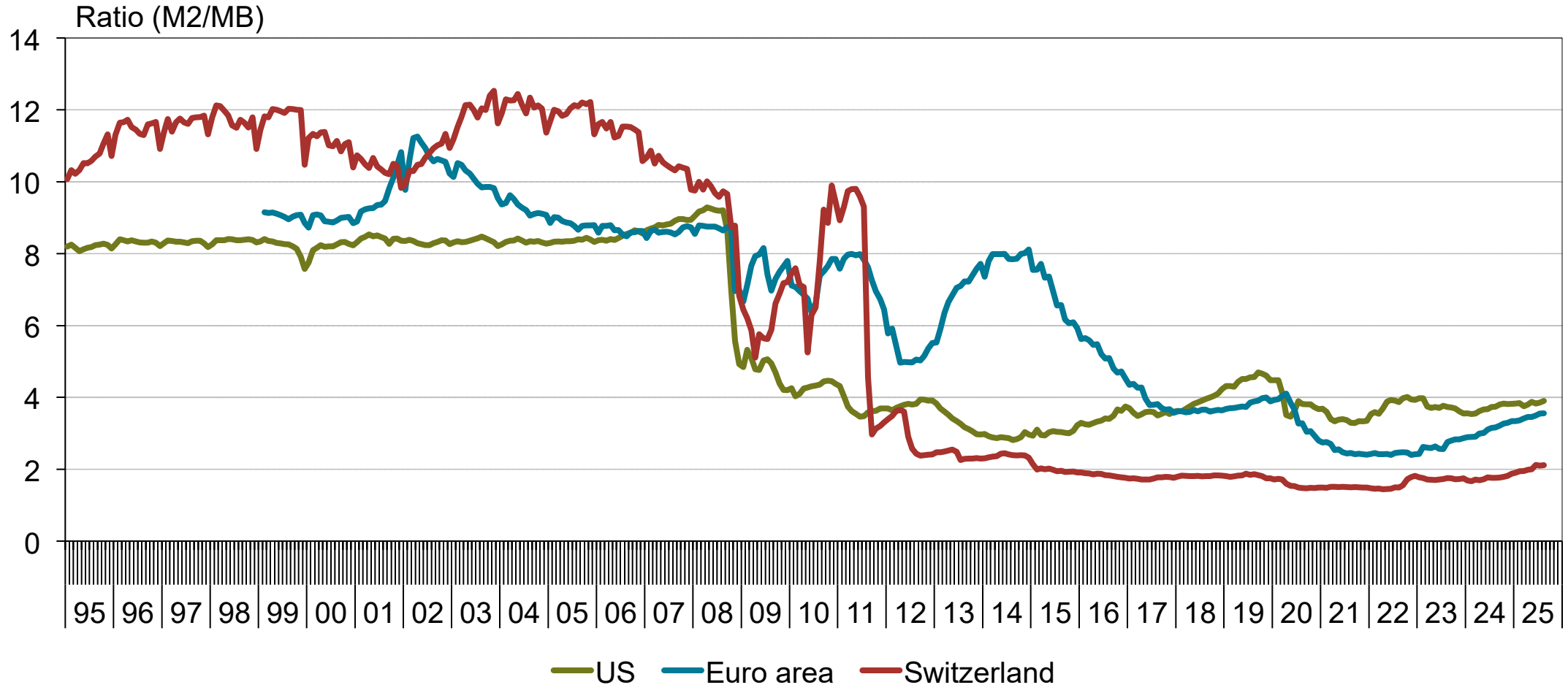
Source: ECB

Developments of different concepts of money: Monetary Base vs. M2



Sources: Fed, BoJ, ECB, SNB

Developments of the money multiplier (M2 / Monetary Base)



Sources: Fed, BoJ, ECB, SNB

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Principles of Macroeconomics

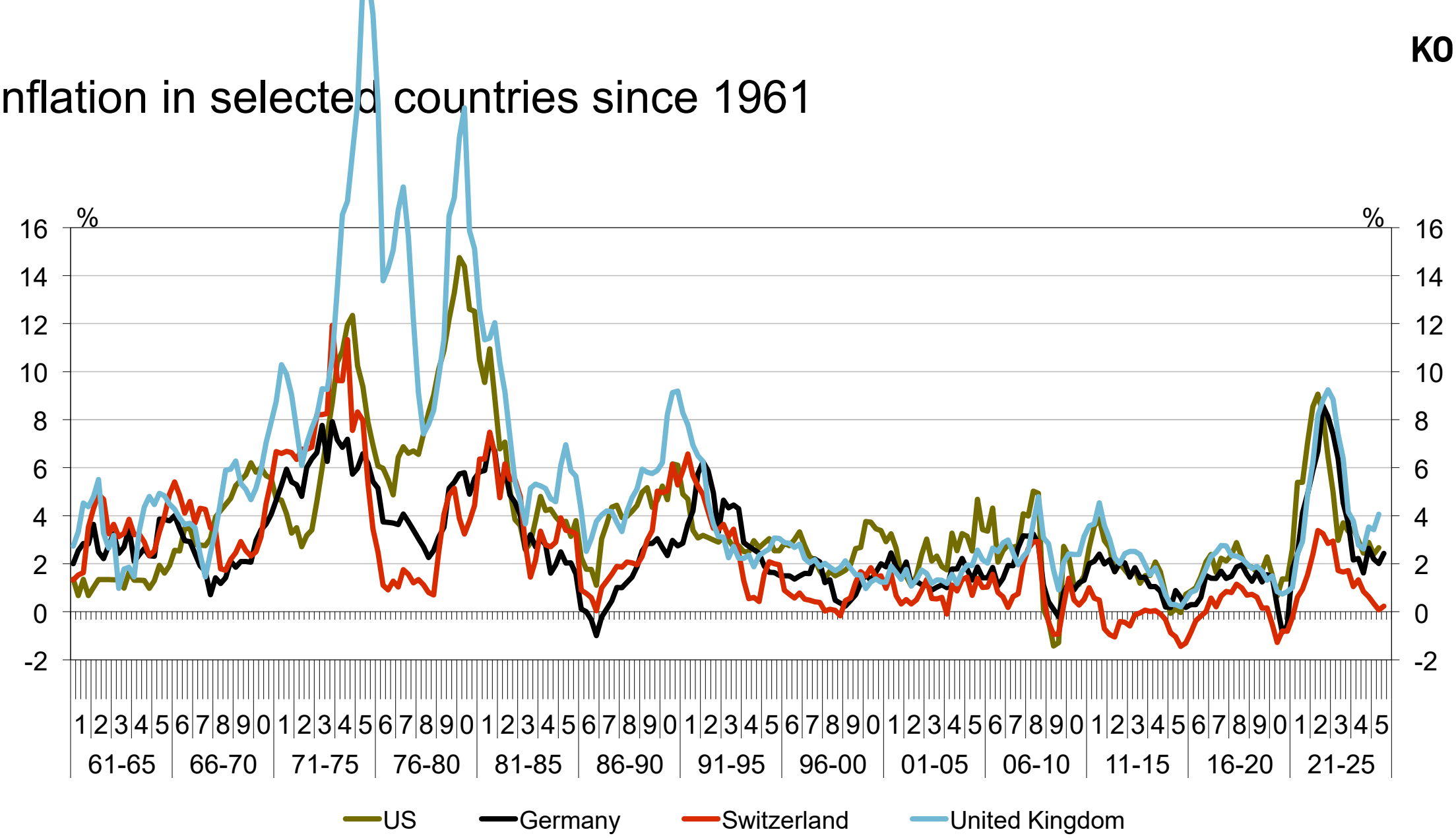
Lecture 6: The monetary system

Prof. Dr. Jan-Egbert Sturm

The Classical Theory of Inflation

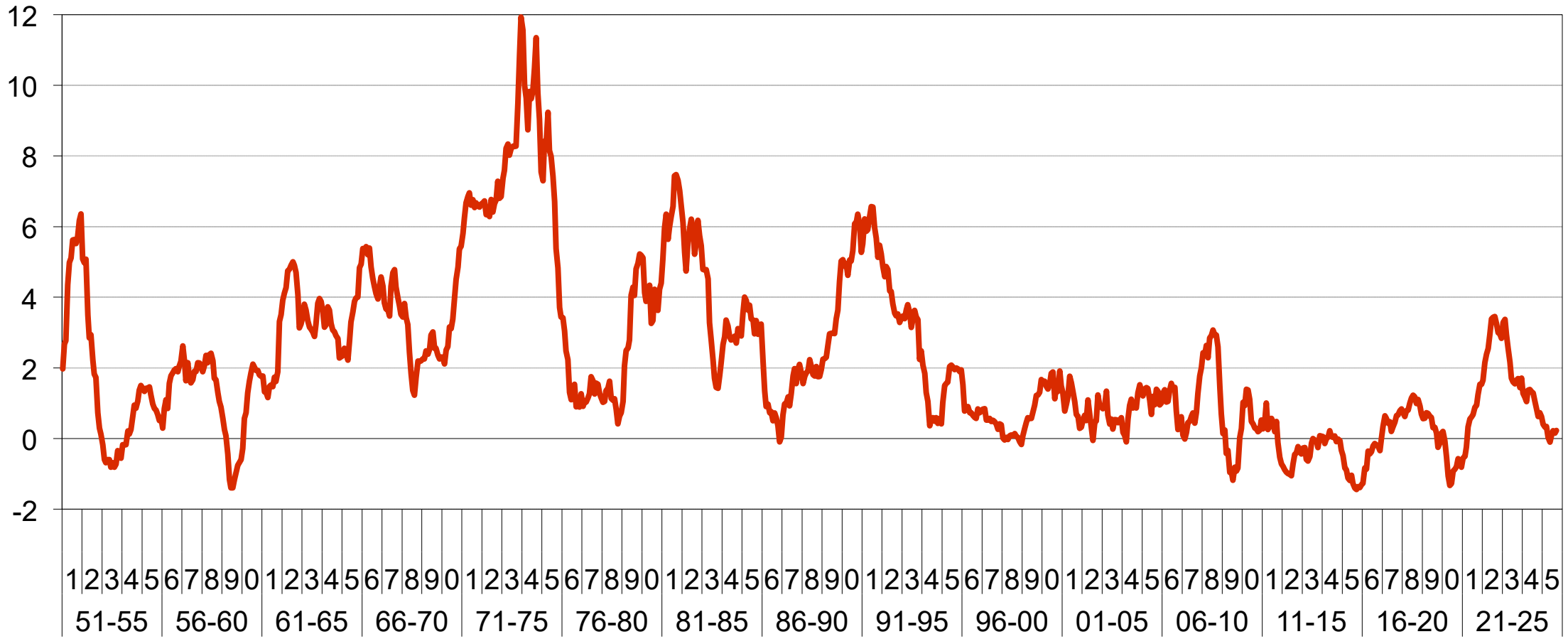
- Inflation is an increase in the overall level of prices
  - In the 1970s prices in the U.S. rose by 7 % per year
  - During the 1990s, prices in the U.S. rose on average by 2 percent per year
  - Inflation in the UK exceeded 20% per year in the mid-1970s
  - In the late 1990s and early 2000s UK inflation has been low and stable at around 2% per year
  - In the 1970s the average inflation rate was around 5% in Switzerland
  - In the 2010s the average inflation rate was around 0% in Switzerland
- Hyperinflation is an extraordinarily high rate of inflation
  - In Venezuela inflation was 929 800% in 2018
- Deflation, meaning decreasing average prices, occurred regularly in the nineteenth century…
  - and in Switzerland during 2012-2016 (with an average inflation rate of -0.5%)

Inflation in selected countries since 1961



Sources: OECD, BIS, Eurostat

Inflation in Switzerland since 1951



Source: SFSO

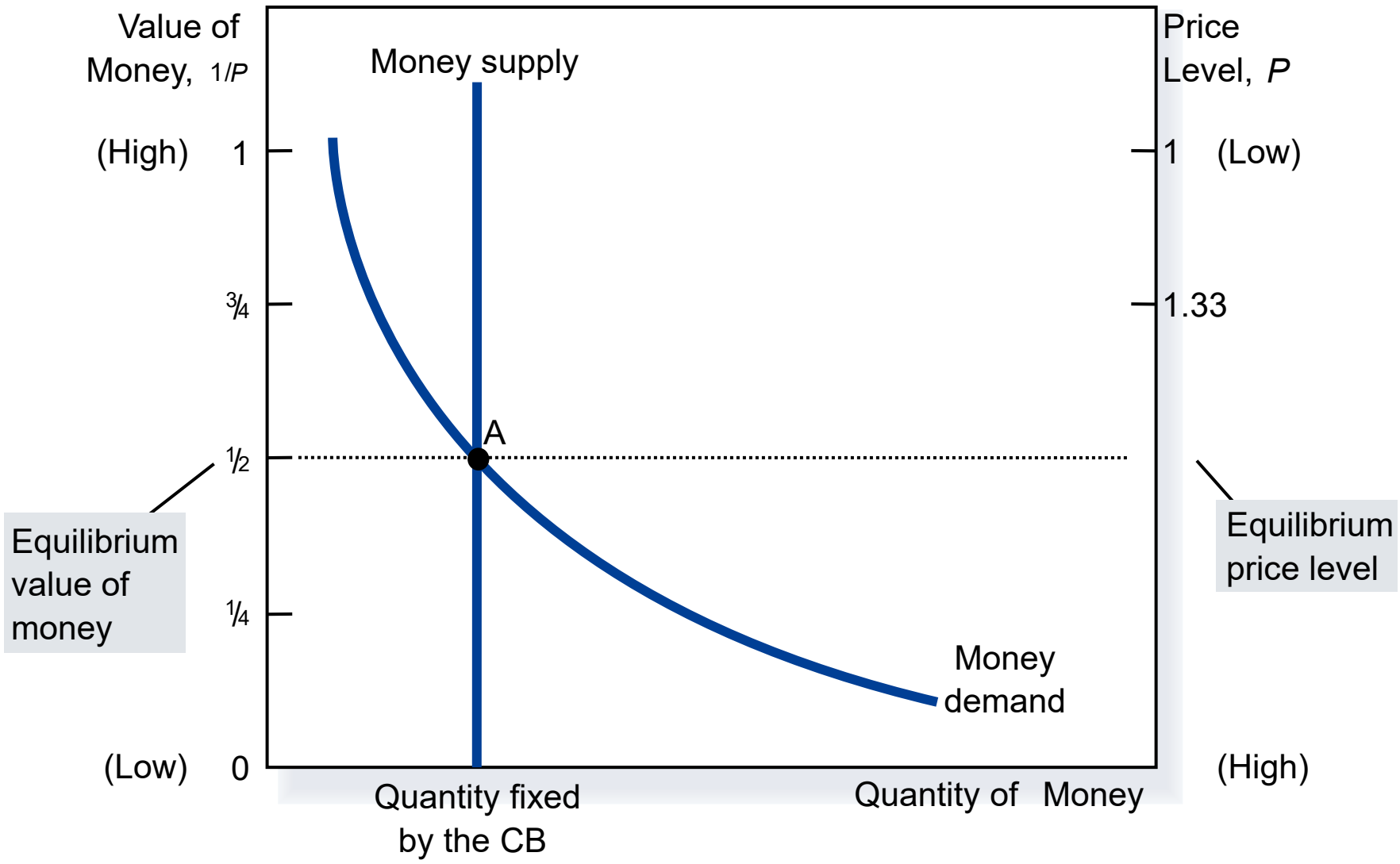
The Level of Prices and the Value of Money

- The quantity theory of money is used to explain the long-run determinants of the price level and the inflation rate
- Inflation is an economy-wide phenomenon that concerns the value of the economy's medium of exchange
- When the overall price level rises, the value of money falls

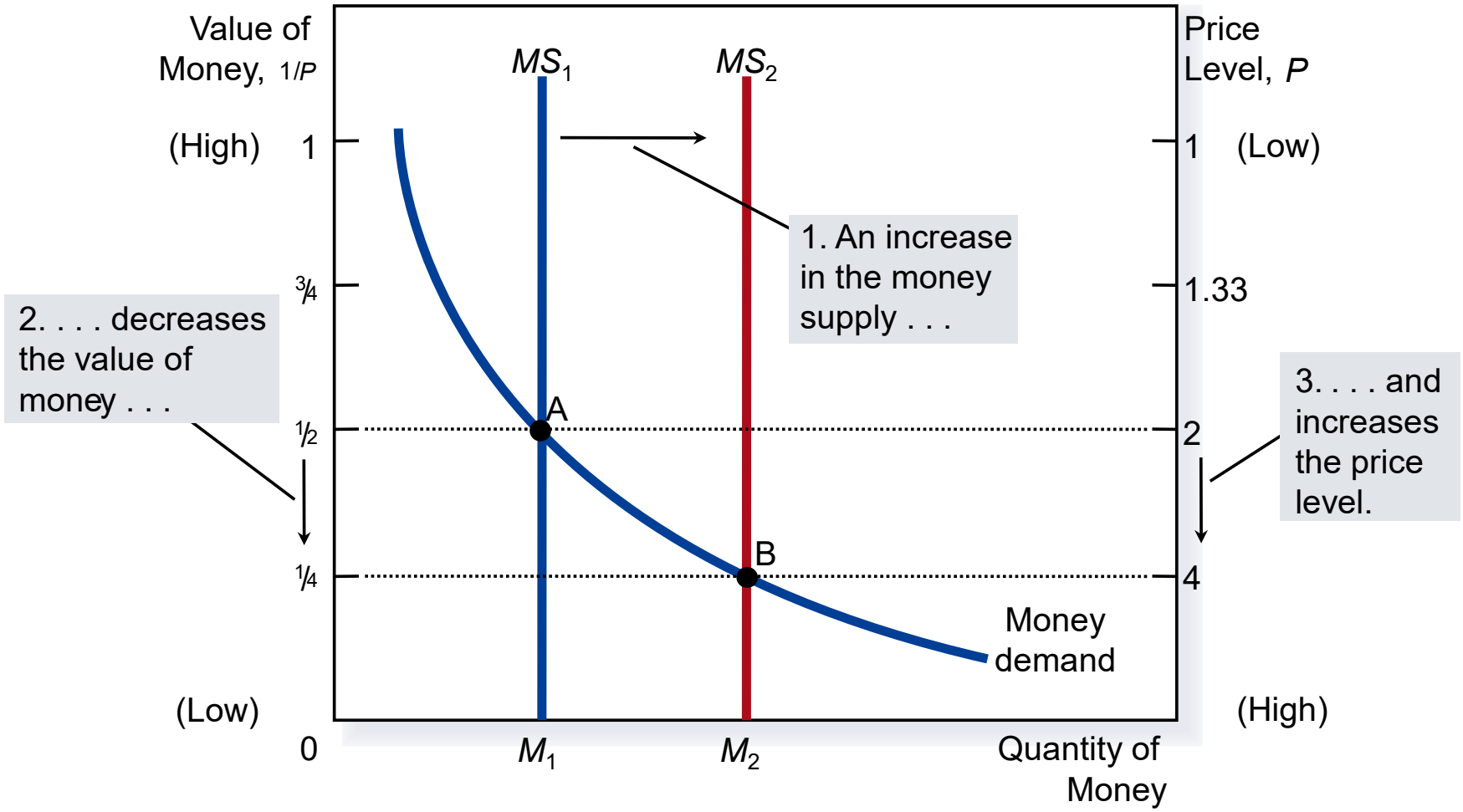
Money Supply, Money Demand, and Monetary Equilibrium

- The value of money is determined by the supply and demand for money
- The ***money supply*** is a policy variable that is controlled by the central bank
  - Through instruments such as open-market operations, the CB affects the quantity of money supplied
  - Therefore the supply of money is (assumed to be) vertical (perfectly inelastic)
- ***Money demand*** has several determinants, including interest rates and the average level of prices in the economy
  - People hold money because it is the medium of exchange
  - The amount of money people choose to hold depends on the prices of goods and services
  - Therefore the lower the value of money the higher the demand
- In the long run, the overall level of prices adjusts to the level at which the demand for money equals the supply

How Supply and Demand for Money Determine the Equilibrium Price



An Increase in the Money Supply



The Effects of a Monetary Injection

- The ***Quantity Theory of Money***
  - A theory explaining how the price level is determined and why it might change over time
    - The quantity of money available in the economy determines the value of money
    - The primary cause of inflation is the growth in the quantity of money

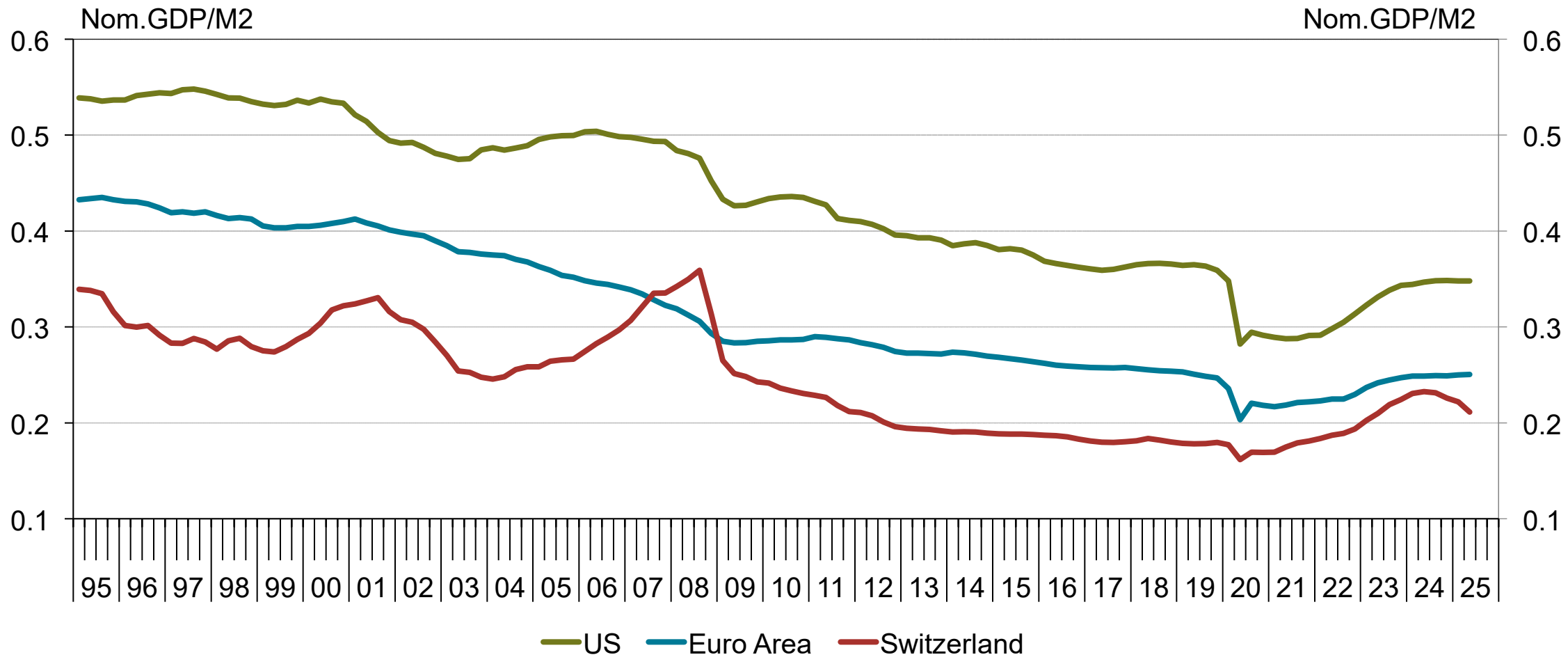
The Classical Dichotomy and Monetary Neutrality

- ***Nominal variables*** are variables measured in monetary units
- ***Real variables*** are variables measured in physical units
- In the long run, real economic variables do not change with changes in the money supply
  - According to the ***classical dichotomy***, different forces influence real and nominal variables
- Changes in the money supply affect nominal variables but not real variables
- The irrelevance of monetary changes for real variables is called ***monetary neutrality***

Velocity and the Quantity Equation

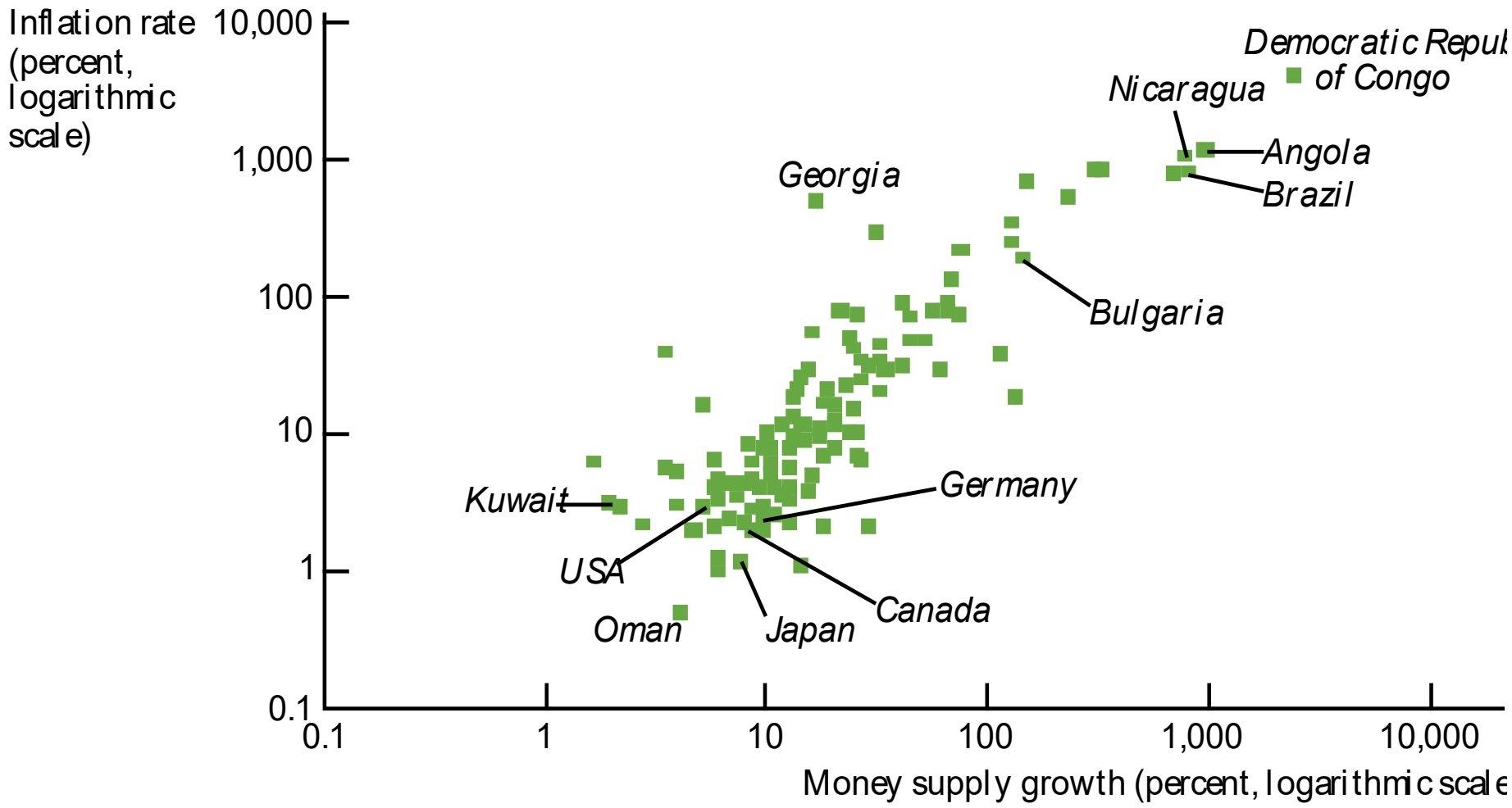
- The ***velocity of money*** refers to the speed at which money (CHF/€/$) travels around the economy:
  $V \equiv \frac{P \times Y}{M}$, where V = velocity, P = price level, Y = quantity of output, M = quantity of money
- Rewriting the equation gives the ***quantity equation***: $M \times V = P \times Y$
  - The quantity equation relates the quantity of money (M) to the nominal value of output ($P \times Y$).
- The quantity equation shows that an increase in the quantity of money in an economy must be reflected in one of three other variables:
  - The price level (P) must rise / the quantity of output (Y) must rise / the velocity of money (V) must fall
- The ***Quantity Theory of Money***
  - The velocity of money (V) is relatively stable over time
  - When money (M) changes, it causes proportionate changes in the nominal value of output ($P \times Y$)
  - Because money is (in the long-run) neutral (Y is determined on goods and service markets), money (M) does not affect output (Y)

The (quarterly) velocity of money (M2)

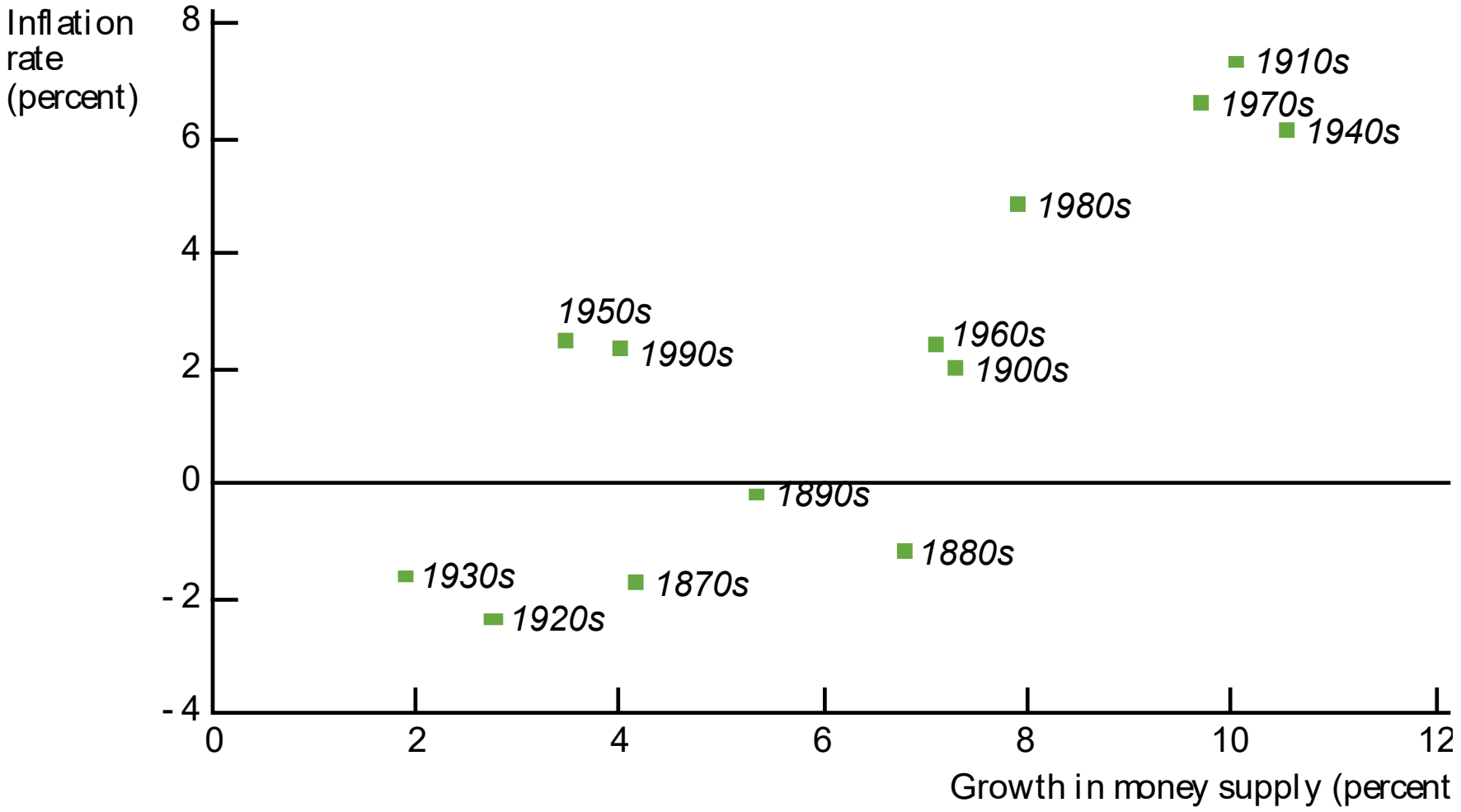


Sources: Fed, BoJ, ECB, SNB, BEA, Eurostat, Seco

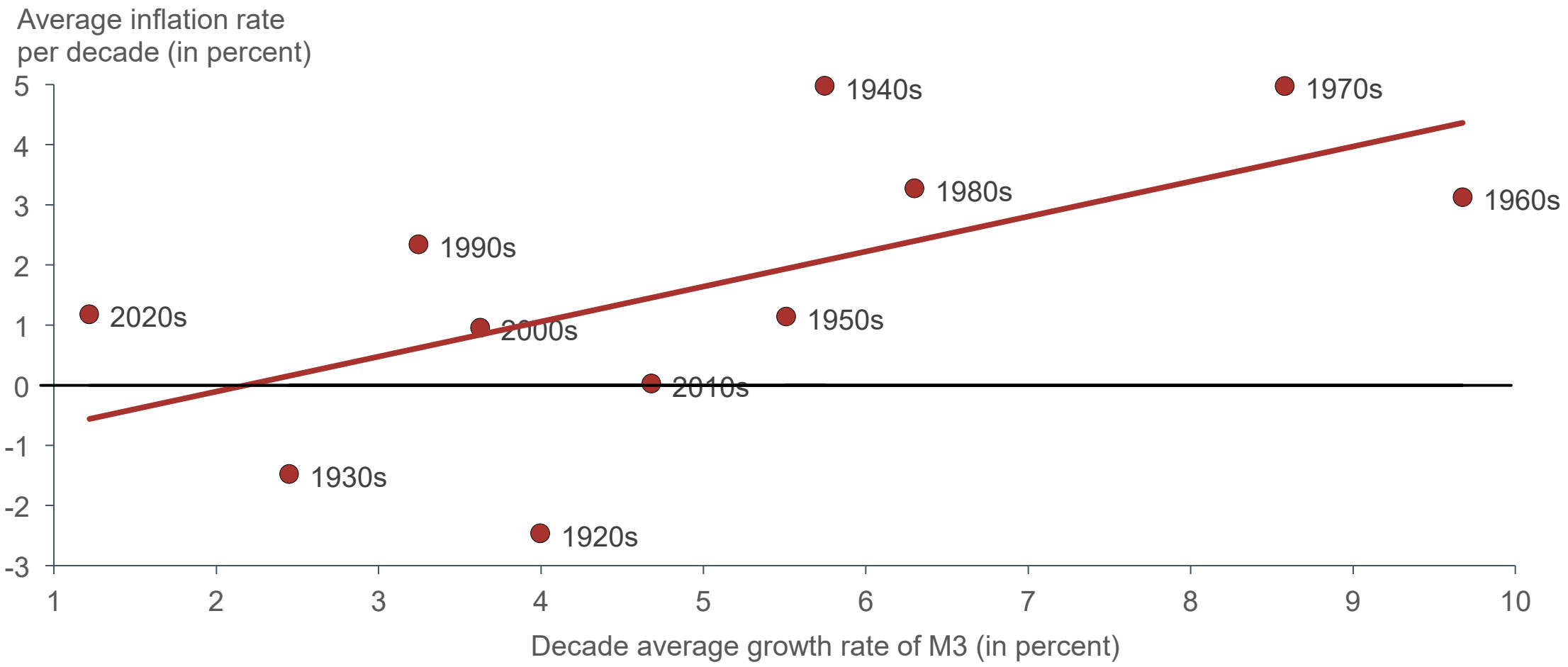
International data on inflation and money growth



U.S. data on inflation and money growth



Swiss data on inflation and money growth

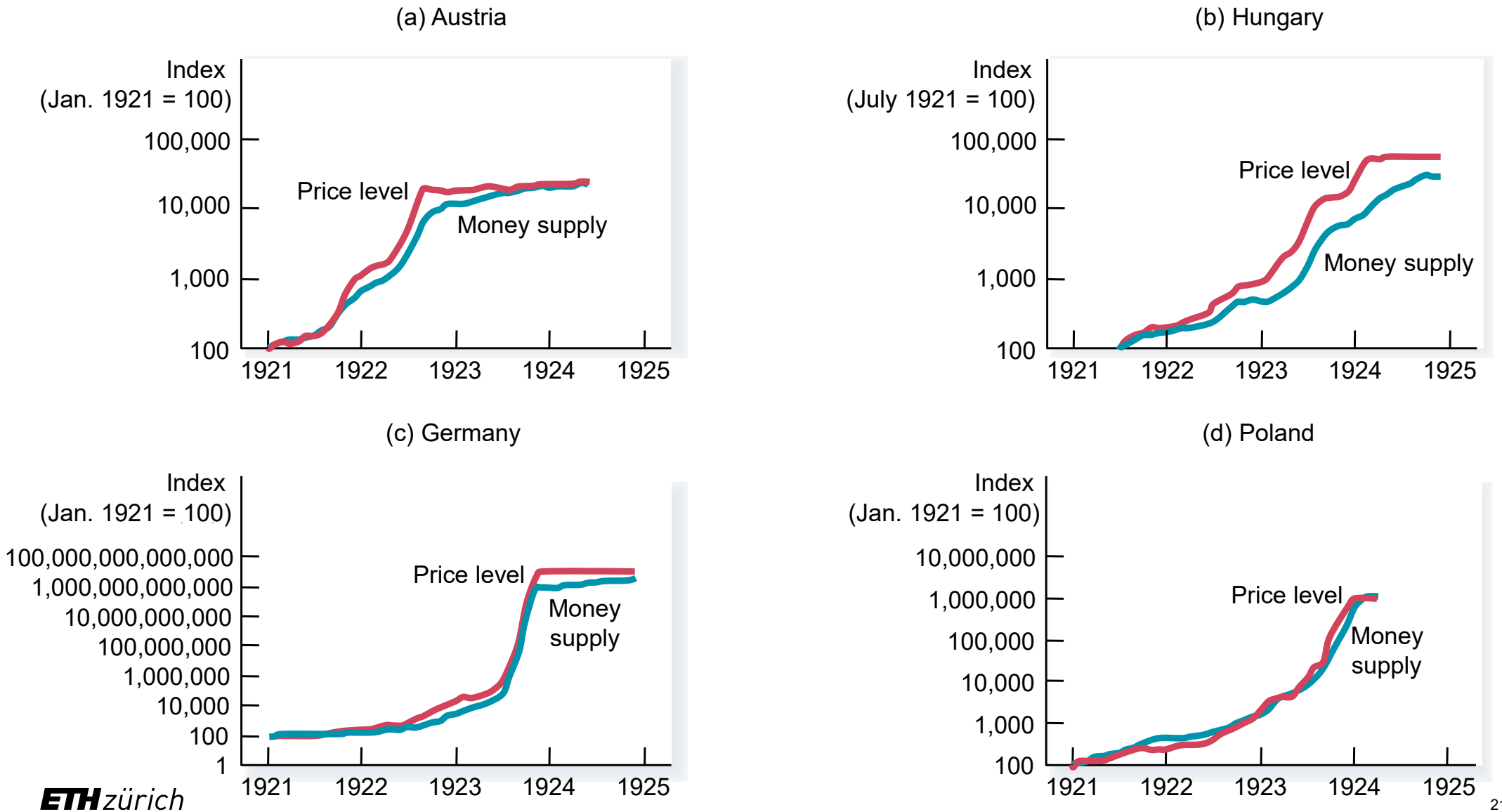


Sources: SNB, SFSO, Grüebler

Case Study: Money and Prices during Four Hyperinflations

- Hyperinflation is sometimes defined as inflation that exceeds 50 percent per month
- Hyperinflation often occurs because the government prints too much money to pay for its spending

Money and Prices During Four Hyperinflations in the 1920s



More recent example: Zimbabwe (The Guardian, Friday 12 June 2015)

- Zimbabwe offers new exchange rate: $1 for 35,000,000,000,000,000 old dollars
- Central bank discards local currency after years of hyperinflation which at one point reached 500,000,000,000%
  - The southern African country started using foreign currencies including the US dollar and South African rand in 2009 after the Zimbabwean dollar was ruined by hyperinflation, which hit 500 billion per cent in 2008
  - http://www.theguardian.com/world/2015/jun/12/zimbabwe-offers-new-exchange-rate-1-for-35000000000000000-old-dollars
- Even more recent example: Venezuela, 2018>
  - https://mobile.reuters.com/article/amp/idUSKBN1KD2L9
  - In 2019, the CPI increased by more than 65,000%
  - By 2020, it had settled down to a “mere” 2,360%

The Inflation Tax

- When the government raises revenue by printing money, it is said to levy an inflation tax
- An ***inflation tax*** is like a tax on everyone who holds money
- The inflation ends when the government institutes fiscal reforms such as cuts in government spending

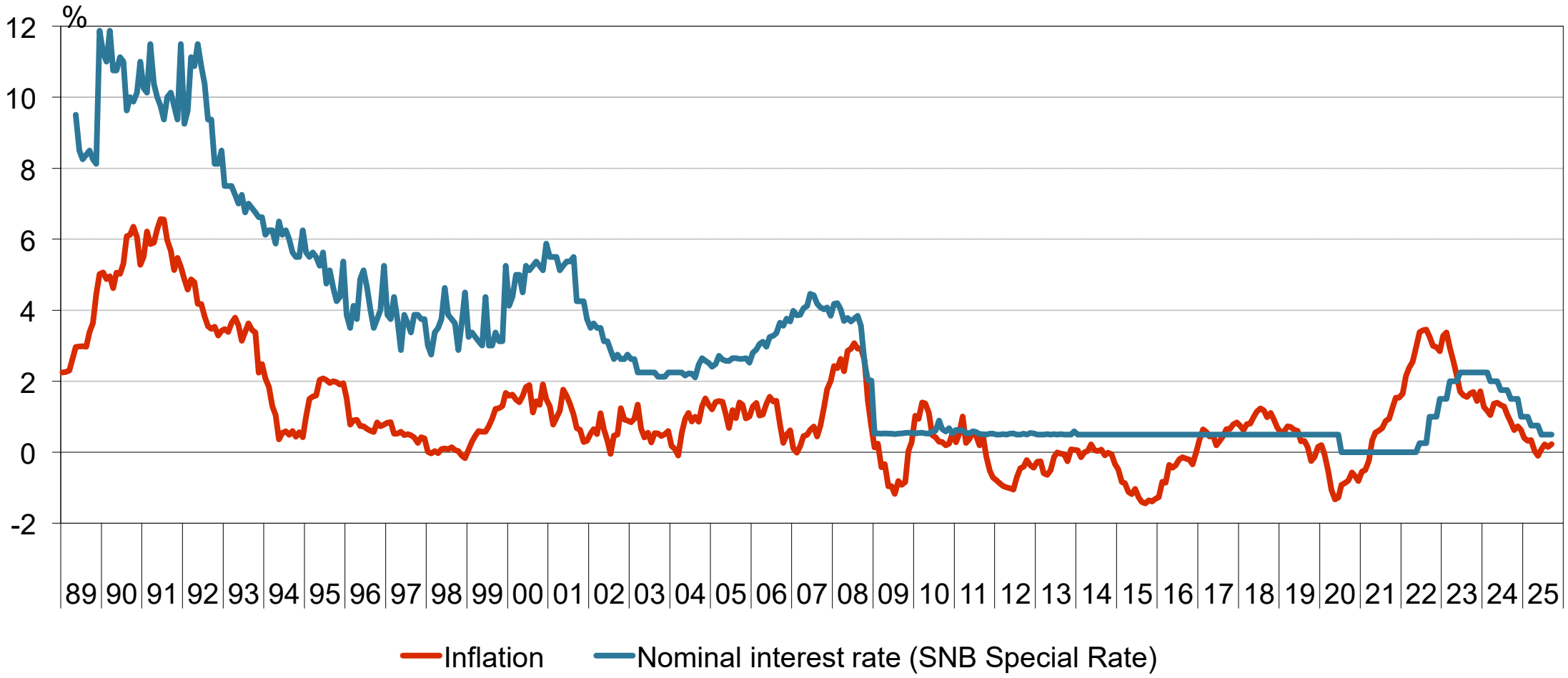
The Fisher Effect

- Remember: nominal interest rate = real interest rate + inflation rate

$$i = r + \pi$$

- The ***Fisher effect*** refers to a one-to-one adjustment of the nominal interest rate to the inflation rate
- According to the Fisher effect, when the rate of inflation rises, the nominal interest rate rises by the same amount
- The real interest rate stays the same
  - It is determined on the loanable funds market (real S = real I)

Nominal Interest and Inflation Rate in Switzerland

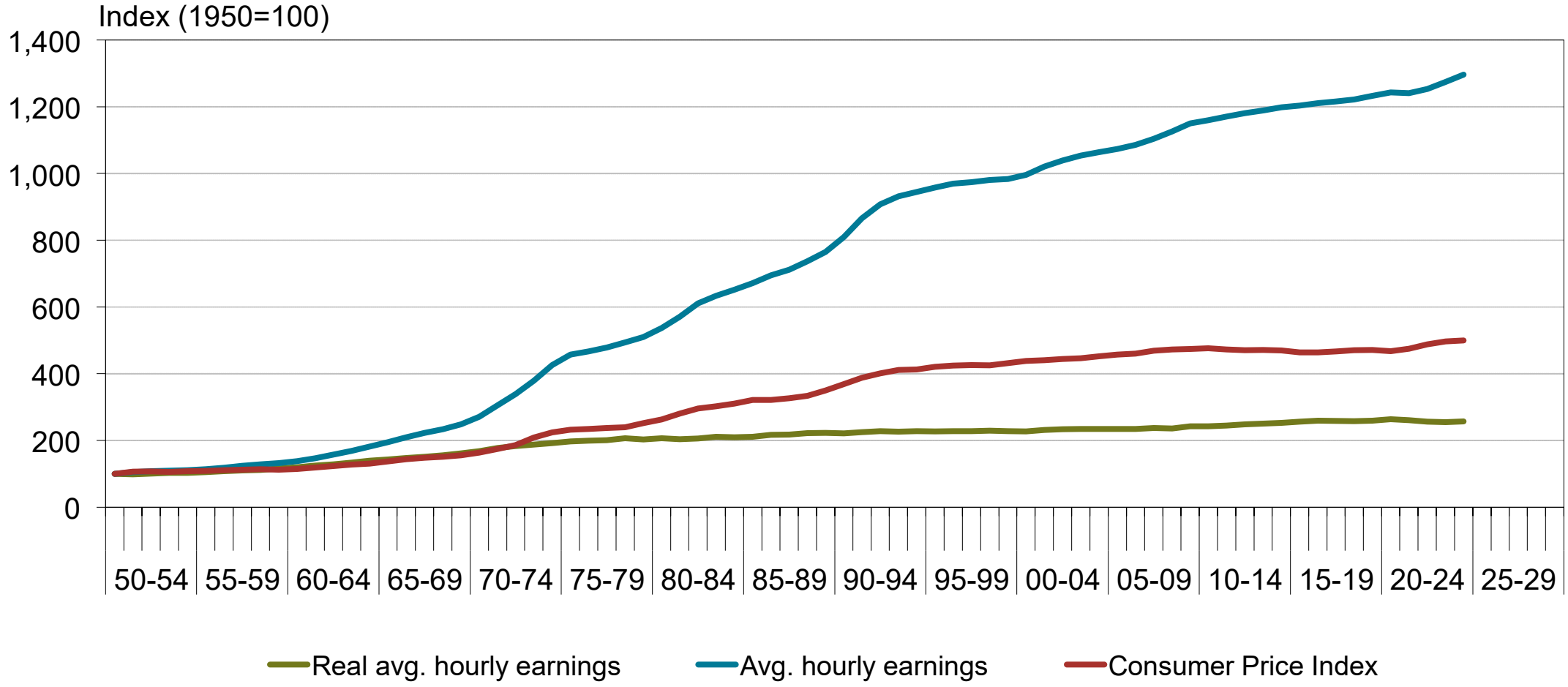


Sources: SFSO, SNB

What are the Costs of Inflation?

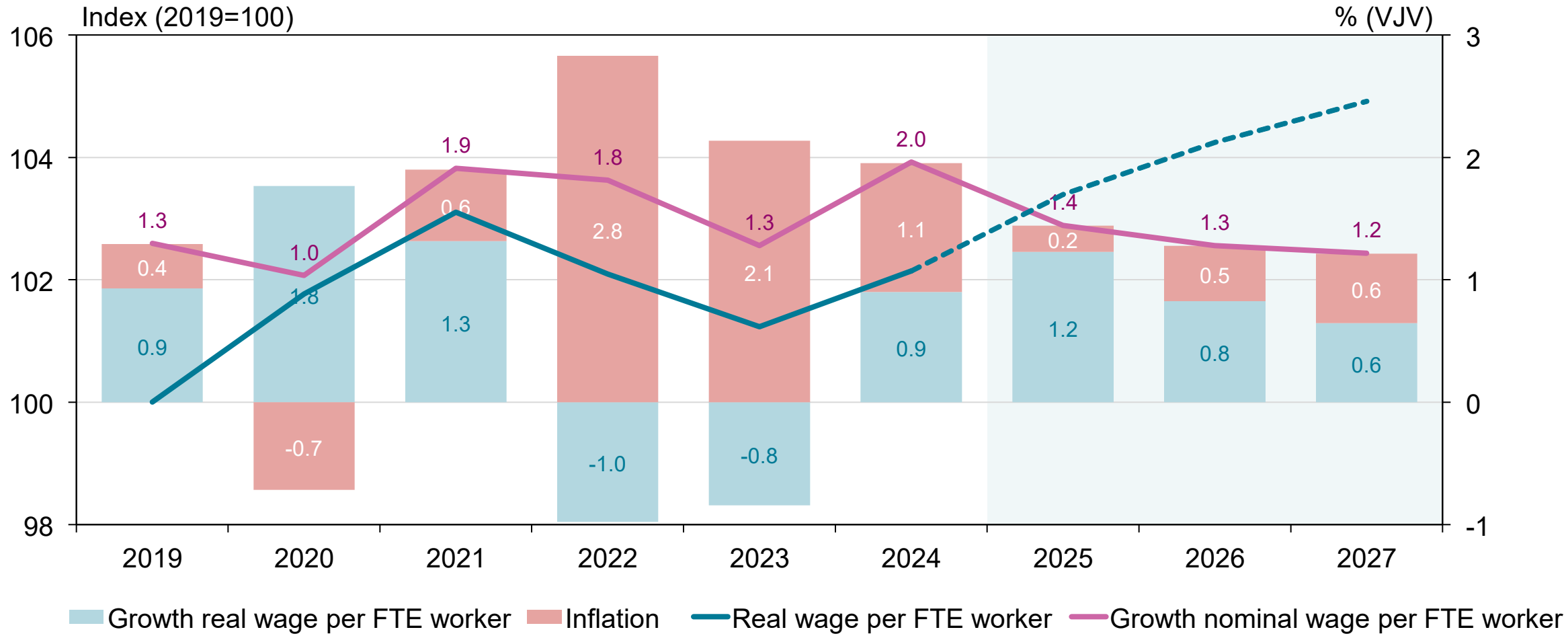
- A Fall in Purchasing Power?
- Inflation does not in itself reduce people's real purchasing power
- As prices rise, so do incomes
  - Inflation does not in itself reduce the purchasing power of incomes
- However, not everyone's income rises exactly with inflation
- Common **misperception**:
  - ***inflation reduces real wages***
  - This is true only in the short run, when nominal wages are fixed by contracts
  - In the long run, real wages are determined by labour supply and the marginal product of labour, not the price level or the inflation rate

Average hourly earnings & the Consumer Price Index in Switzerland



Source: SFSO

Wage development in Switzerland – a short-run perspective



Sources: SECO, KOF (16.06.2025)

The Costs of Inflation

1. Shoeleather costs
2. Menu costs
3. Relative price variability
4. Tax distortions
5. Confusion and inconvenience
6. Arbitrary redistribution of wealth

1. Shoeleather Costs

- ***Shoeleather costs*** are the resources wasted when inflation encourages people to reduce their money holdings
- Inflation reduces the real value of money, so people have an incentive to minimize their cash holdings
- Less cash requires more frequent trips to the bank to withdraw money from interest-bearing accounts
- The actual cost of reducing your money holdings is the time and convenience you must sacrifice to keep less money on hand
- Also, extra trips to the bank take time away from productive activities

2. Menu Costs

- ***Menu costs*** are the costs of adjusting prices
- During inflationary times, it is necessary to update price lists and other posted prices
- This is a resource-consuming process that takes away from other productive activities

3. Relative-Price Variability and the Misallocation of Resources

- Inflation distorts relative prices
- If prices of some goods are “sticky” (e.g. because of menu costs) relative to other goods, then inflation is creating changes in relative prices that lead to decisions of market participants that are economically inefficient in the sense that it deviates from the optimal allocation of resources
- Hence, markets are less able to allocate resources to their best use

4. Inflation-Induced Tax Distortion

- Inflation exaggerates the size of capital gains and increases the tax burden on this type of income
- With progressive taxation, capital gains are taxed more heavily
- The income tax treats the nominal interest earned on savings as income, even though part of the nominal interest rate merely compensates for inflation
- The after-tax real interest rate falls, making saving less attractive

How Inflation Raises the Tax Burden on Saving

	Economy A (price stability)	Economy B (inflation)
Real interest rate	4%	4%
Inflation rate	0	8
Nominal interest rate (real interest rate + inflation rate)	4	12
Reduced interest due to 25 percent tax (.25 × nominal interest rate)	1	3
After-tax nominal interest rate (.75 × nominal interest rate)	3	9
After-tax real interest rate (after-tax nominal interest rate − inflation rate)	3	1

5. Confusion and Inconvenience

- When the Central Bank increases the money supply and creates inflation, it erodes the real value of the unit of account
- Inflation causes dollars at different times to have different real values
- Therefore, with rising prices, it is more difficult to compare real revenues, costs, and profits over time

A Special Cost of ***Unexpected*** Inflation:

6. Arbitrary Redistribution of Wealth

- Unexpected inflation redistributes wealth among the population in a way that has nothing to do with either merit or need
- These redistributions occur because many loans in the economy are specified in terms of the unit of account—money

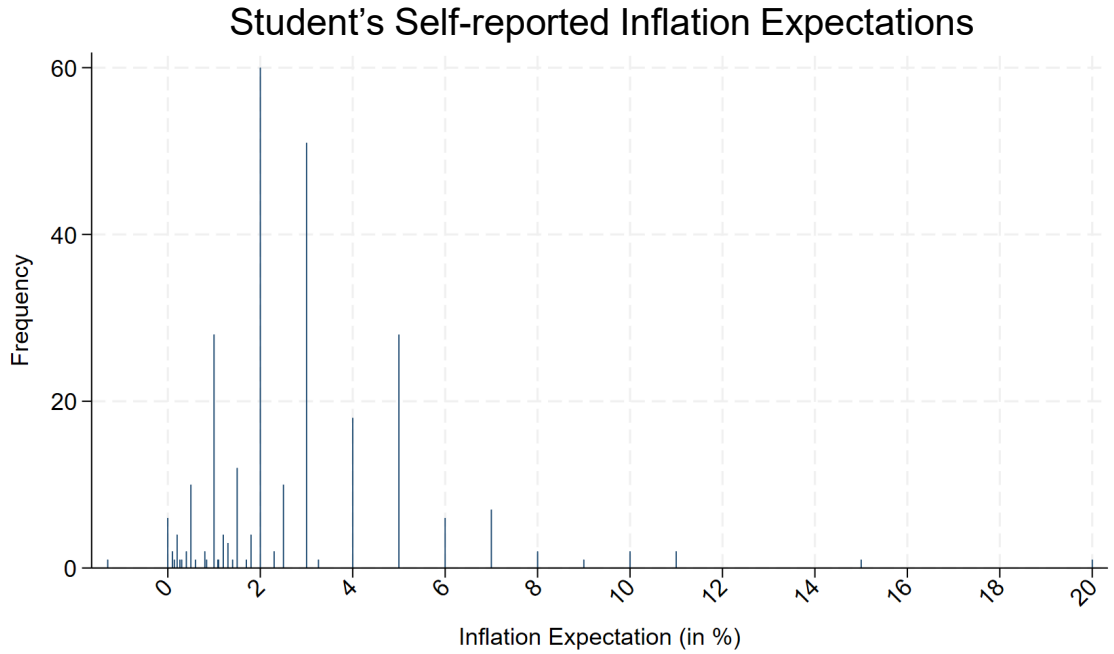
Deflation

- ***Deflation*** where the price level actually falls
- Deflation can be as damaging as inflation because:
  - There is little incentive to spend today if the expectation is for cheaper prices tomorrow
  - It might result in consumers not spending at levels that provide incentives for firms to invest in new capacity
  - little or no growth and with that….
    - increased likelihood of unemployment

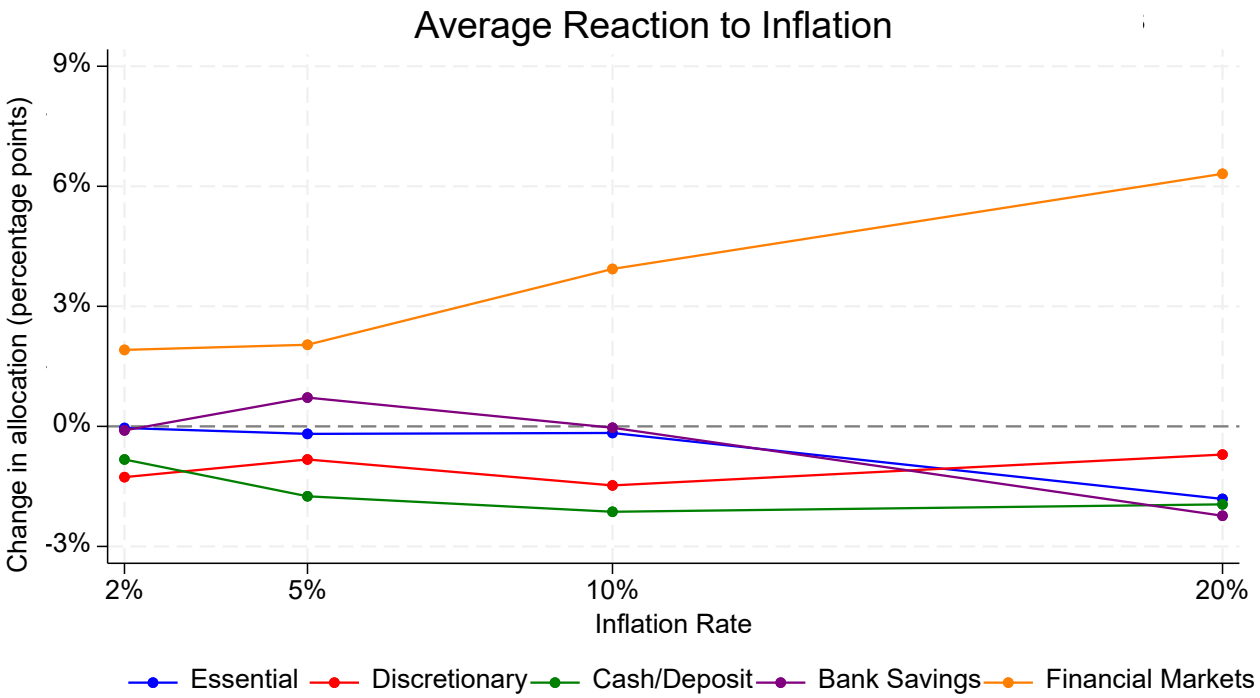
How Should One React to Inflation? Survey from Lecture 2

Inflation Expectations and Expected Stock Market Returns

The students' median expected inflation in Switzerland is 2% for the next year.



Students' income allocation patterns vary as a function of the inflation level they are informed about.



Note: 50.5% of students indicated that they invest in the stock market.

Your Answers Depend on…

…. Factors like Risk Aversion

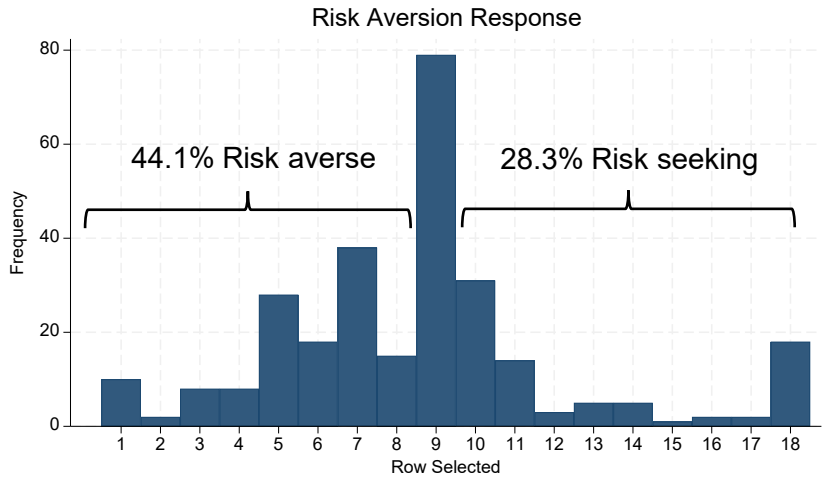
Choose between:

	Sure Payout	Lottery
1	100 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
2	150 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
3	200 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
4	250 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
5	300 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
6	350 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
7	400 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
8	450 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
9	500 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
10	550 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
11	600 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
12	650 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
13	700 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
14	750 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
15	800 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
16	850 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
17	900 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF
18	950 CHF for sure	50% chance to win 1’000 CHF and 50% chance to win 0 CHF

A risk-neutral agent calculates the *expected value of the payoff*:

E(payoff) = 0.5 x 0 + 0.5 x 1’000 CHF = 500 CHF

A risk-neutral agent becomes indifferent in row 9. Most people however are *risk averse* and cannot afford to lose money. Risk aversion can vary depending on an individual’s wealth and other personal factors.



Summary

- The overall level of prices in an economy adjusts to bring money supply and money demand into balance.
- When the central bank increases the supply of money, it causes the price level to rise.
- Persistent growth in the quantity of money supplied leads to continuing inflation.
- The principle of money neutrality asserts that changes in the quantity of money influence nominal variables but not real variables.
- A government can pay for its spending simply by printing more money.
- This can result in an “inflation tax” and hyperinflation.
- According to the Fisher effect, when the inflation rate rises, the nominal interest rate rises by the same amount, and the real interest rate stays the same.
- Many people think that inflation makes them poorer because it raises the cost of what they buy.
- This view is a fallacy because inflation also raises nominal incomes.

Summary

- Economists have identified six costs of inflation:
  1. Shoeleather costs
  2. Menu costs
  3. Increased variability of relative prices
  4. Unintended tax liability changes
  5. Confusion and inconvenience
  6. Arbitrary redistributions of wealth